Estimator help guide

February 2024

EY
Building a better working world

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1 Overview

1.1 Purpose and audience

The Estimator tool's purpose is to enable the user to estimate work effort accurately and efficiently, matching the scope of a project, entire engagement or an enhancement. It also enables creating a resource plan that matches and aligns with the Estimator's effort in terms of effort and properly skilled resources. The primary users of the tool are the solution architects who are working on proposal in the context of the pursuit process and project managers managing delivery. Other common users are executives, pursuit teams, engagement managers, client executives and technical leads. The Estimator will assist users in a proposal process, prior to entering data into Mercury to establish a pricing plan, preparing for deal governance or deal reviews within the sales process.

Other scenarios where the Estimator is used would be in an existing project when new scope is introduced, or users are re-estimating a new part of a project.

This document will describe various elements of the Estimator for the end user. It is intended to be an all-inclusive help guide.

For first-time users, starting by trying a basic estimation will help get accustomed to the Estimator and the process.

Advanced users will want to review the appendix of this document to find the finer points of the Estimator, which will contain any nuances, edge cases or limitations.

1.2 User experience, standards, conventions and benefits

User experience is a fundamental objective. The Estimator contains visual elements, such as colors, fonts, pop-up descriptions and worksheet ordering, to help the user navigate. The standard conventions used in the Estimator include:

- Worksheets that guide the user through the process
- Blue cells representing editable areas
- Grey and darker colored cells representing fixed data
- Slightly larger font size to highlight the summary tables
- Purple font to denote fixed estimator data
- Standard borders and frames around different sections to help compartmentalize the different sections of the Estimator
- Summary reference and staffing dashboards on key worksheets that reduce the need to flip between worksheets
- Selected items explained via pop-ups when hovered over, which are also available in the glossary (explained as part of this document)

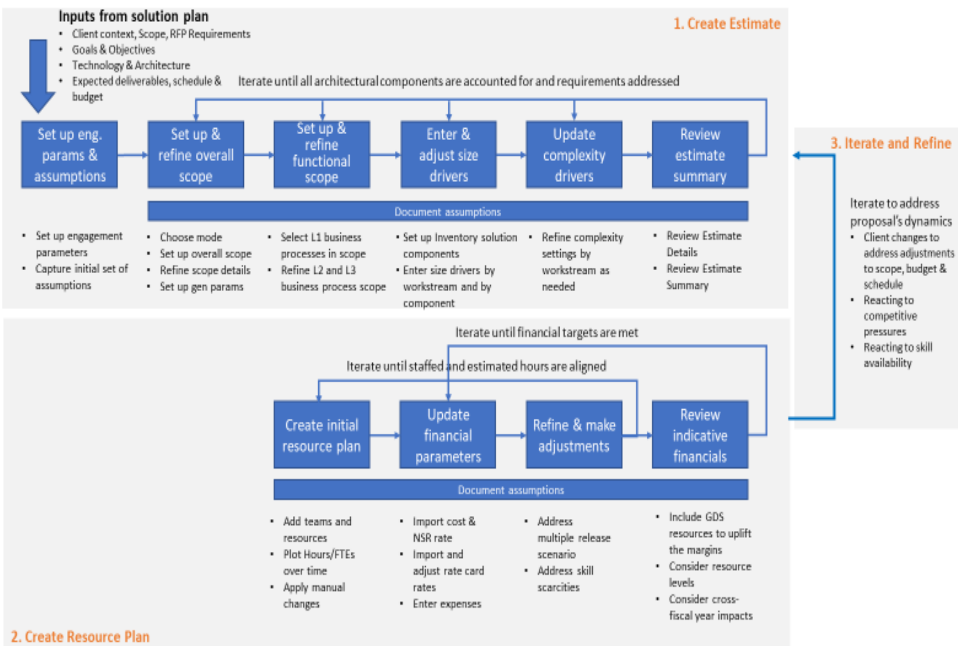
User benefits:

- Standard estimating approach for a given domain/subdomain
- A framework for encoding an estimating approach that can be applied with discipline and consistency
- Aligns with the Digital Delivery Methodology on key concepts such as phases and stages of work, roles definitions, and linkage to the standard agile process

- Provides common size and complexity drivers definitions that can be included into SOW and serve as a baseline for managing scope during the delivery
- Provides immediate feedback on impact on key indicative financial metrics as changes to the estimate and/or resource plan are applied:
    - These are considered "indicative" because they do not automatically match what the Mercury pricing plan will calculate. However, they should be very close and an indicator of the Mercury calculation.

1.3 Estimating process flow

This diagram illustrates the major steps in making estimates and creating a staffing plan. A bulleted list of items associated with each box in the flow diagram describes what happens in each step with a forward reference for more details.



The rest of this document is structured in a way that aligns with the process above, organized into three major sections:

- Create estimate
- Create resource plan
- Review and refine

Each major section covers the following topics:

- **Overview** – the functional description of each step
- **Worksheet details** – describing worksheet functionality on each worksheet

- **Using this worksheet** — details for how to accomplish a given task and linkage to the worksheet functionality

At the end of the document, an appendix is included to address several special topics. These topics are forward referenced throughout the document.

2 Create effort estimate

Creating an effort estimate involves the following steps. Each is described in detail in the sections that follow:

- Set up engagement parameters and assumptions
- Set up and refine scope of the estimate
- Set up and refine functional scope
- Enter and adjust size drivers
- Update complexity drivers
- Review estimate summary

2.1 Set up engagement parameters and assumptions

2.1.1 Overview

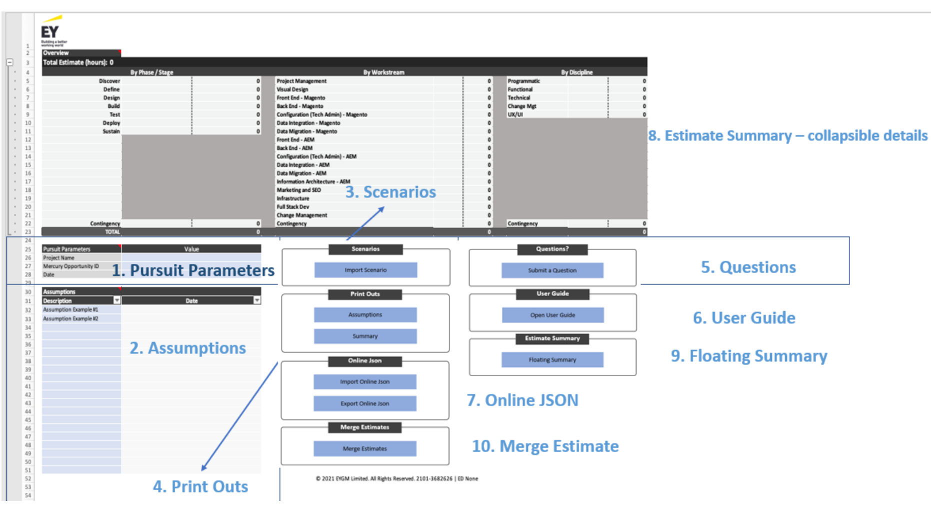
In this step, users capture the pursuit parameters for the deal, RFP or project being estimated and document the assumptions based on which estimate and resource plans are established.

Users should update the assumptions as they progress through the estimating and resource planning steps. Capturing such assumptions on the Assumption worksheet is critical to supporting the estimate. The rationale behind the assumption section is twofold:

- Provide a context for evaluating the accuracy of the estimate when reviewed by the technical leadership participating in the pursuit process
- Enable quick and efficient capture of the key assumptions within the Digital Delivery Methods (DDM) Solution Plan template and eventually in the statement of work (SOW)

Advanced users may enhance the assumptions by summing up and linking to other data within the Estimator.

2.1.2 Start worksheet details



This worksheet contains engagement parameters as well as assumptions that are most crucial to the project. It also provides a sample estimate summary based on workstreams, stages and disciplines:

- **Pursuit Parameters** – list of parameters
- **Assumptions** – list of assumptions listed with brief description
- **Scenarios** - pre-configured scenario that user can import to build a new estimate
- **Print Outs** - user can print out summary of scope table and objects as well as any parameters that are overridden
- **Questions** - ask any questions to the estimator team
- **User Guide** - this book that highlights the features of the estimator in details
- **Online JSON** - a new feature where users can import an online estimator into this excel estimator and export the excel estimator into online platform
- **Estimate Summary – collapsible details** – summary of total hours estimated for workstreams, stages and disciplines
- **Floating Summary** - a summary of total hours estimated for workstreams, stages and disciplines as well of resourcing hours and indicative financials (if available for estimate) in a way of floating window over the excel.

2.1.2.1 Pursuit Parameters

Pursuit Parameters	Value
Project Name	
Mercury Opportunity ID	
Date	

Pursuit Parameters – A set of conditions that are typically captured at the start of the estimating process. They often consist of the following:

- **Project name** – name of the engagement
- **Mercury opportunity ID** – billable engagement ID in Mercury
- **Date** – when the parameters are written down

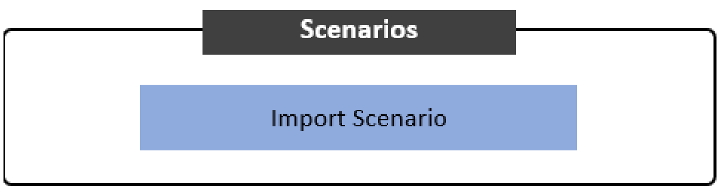
2.1.2.2 Assumptions

Assumptions	
Description	Date
Assumption Example #1	12/12/2021
Assumption Example #2	12/13/2021

Assumptions – set of expectations derived from the context of the proposal or RFP being addressed

- **Description** – a brief description of the assumption
- **Date** – when this assumption was documented

2.1.2.3 Scenarios



A capability to load a setup for different estimating scenario enables the user to get a quick start on the estimating process. The estimating elements such as scope settings, process enabling, component and object inventories can be loaded into the estimator and then tailored to specific deal estimating context. Multiple scenarios differentiated by a specific business context (e.g., Asset Management), specific industry sector or sub-section, and scope - small, medium, and large can be accommodated. Each scenario must be pre-configured in a separately filled scenario template that is available from the SDMT estimator download page.

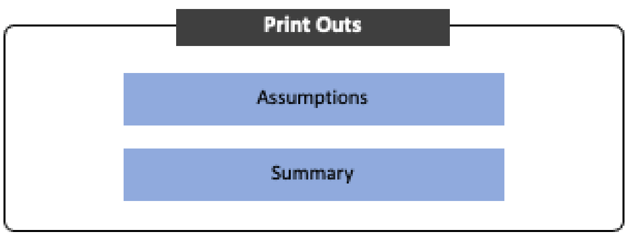
Both Import and Append options are provided. If the Import option is used, then existing component and object inventories are erased before the import is started. The Append operation simply appends new Component and Object inventories to the existing one. Note that Scope, Process hierarchy, and Complexity are not supported by the Append operation - only Components and Objects inventories can be appended to the existing lists.

To initiate the process, the user needs to click on this Import Scenario button and then navigate to one of the pre-filled scenario templates on his/her computer. In the long term, the scenario template will pre-filled by the SMRs within a given practice and normally will not require any updates by the end users. Initially, however, the end user may need to pre-fill, update, or even create new content with the template.

The scenario template consists of Scope, Process, Components and Complexity worksheets. In the scope worksheet of the template, users will be able to change scope of any workstreams, phases, parameters, complexity and size drivers as well as changing any default parameter values. The process sheet deals with process scope selections of components where user is able to change scope status of individual L1, L2 or L3 processes. The components worksheet is where any size, mode or object count of any particular workstream, and its object type can be modified. In the complexity sheet, complexity drivers can be updated for each workstreams. Once any changes are made in the scenario template, users can import one or multiple scenarios into the workbook in 2 ways: a) Override, b) Append.

By clicking on Yes, users can choose to override previously existing workbook and it will copy the exact version of the newly imported scenario. By clicking No, users can append the workbook which will add new updates. In addition, in the object inventory table under Component's worksheet, users can see duplicate values for workstreams, components and the objects with their size, mode and count values. Append will, technically, change the values in complexity sheet as well and import new values.

2.1.2.4 Print Outs



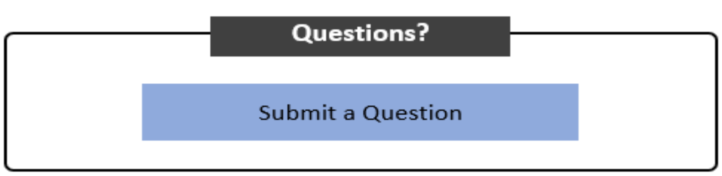
The first button named Assumptions accumulates the following:

- any assumptions in the Start worksheet,
- changes in values or rationale for any parameters in Scope worksheet,
- size drivers whose default value is changed, that is the adjustor is changed from 100% or a new rationale is added
- complexity drivers whose default value is changed under override column

Any changes in the above-mentioned fields gets assembles into a separate worksheet named Assumptions that would be mentioned later this guide.

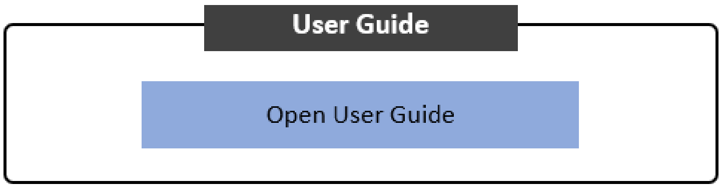
When user clicks on Summary, it prints out a new excel sheet that highlights the overall scope, any objects that were added on in the object inventory table and overall estimate summary by workstreams/phase/discipline. Users are asked whether to include out-of-scope objects and objects with zero count in that summary print out.

2.1.2.5 Questions



This particular macro allows users to send an email to the team regarding any questions/concerns while using the estimator. By clicking on it, it opens an email wizard with the subject line - Questions/Feedback for the Estimator Team.

2.1.2.6 User Guide



This macro will allow users to navigate to this user guide on the EY SharePoint site.

2.1.2.7 Online JSON

Import Online JSON is a new capability that allows user to import JSON export from the online estimator into the Excel estimator. It works exactly as the import scenario where upon importing, user shall be able to identify every component of the online estimator here in the Excel book.

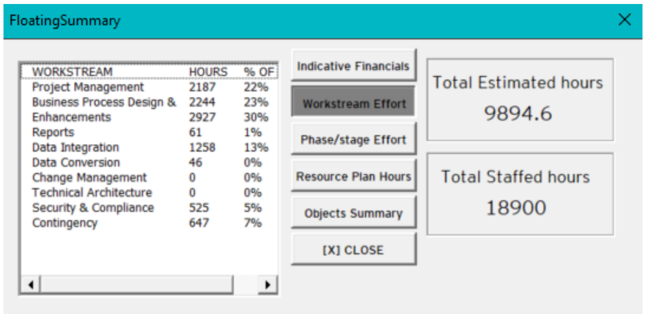
The second button named Export Online JSON does the opposite, that is created a JSON formatted version of this excel estimator that user can import into the online platform.

2.1.2.8 Estimate Summary – collapsible details

Total Estimate (hours): 615.4					
By Phase / Stage		By Workstream		By Discipline	
Discover	0	Project Management	575	Programmatic	575
Plan	108	Business Process Design & Config	0	Functional	0
Confirm & Design	0	Enhancements	0	Technical	0
Progressive Build	208	Reports	0	Change Mgt	0
Test	182	Data Integration	0	UX/UI	0
Deploy	65	Data Conversion	0		
Support	13	Change Management	0		
		Technical Arcitecture	0		
		Security & Compliance	0		
Contingency	40	Contingency	40	Contingency	40
TOTAL	615		615		615

The total effort estimate is always displayed at the top of the worksheet. However, the details normally hidden can be expanded by clicking on a "+" sign. Once expanded, this section contains the estimate summary broken down by phases/stages, workstreams and disciplines. The detailed description of these is identical for all worksheets and can be found under *Estimate*.

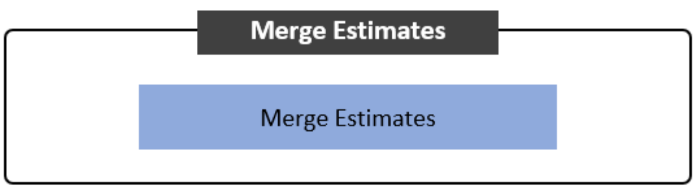
2.1.2.9 Floating Summary



Floating Summary Window enables user to follow on the run Total Estimated Hours for Workstreams, Phases and Disciplines if any applies for the estimate. If applicable, the window can also show Indicative Financials as well as Resource Plan Hours. This feature is made to make it easier to control estimate values while working on specific component without a need to go up to estimate summary or to another tab every time there is a change. **To update values inside Floating Summary Window click on one of the tabs existing inside the Window.**

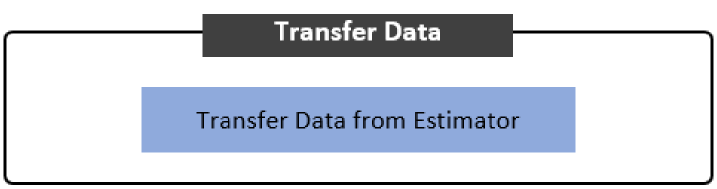
Objects Summary tab – redirects to another floating window called Floating Objects Summary

2.1.2.10 Merge Estimates



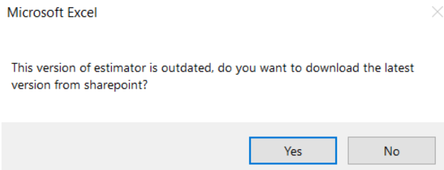
Merge estimates is a feature that allows a user to take two separate Estimators across models if desired and merge their Estimate and Resources into a single model to get an overall summery of the two models in one location.

2.1.2.11 Transfer Data

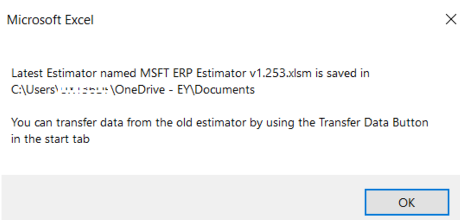


The Transfer data helps the user to update their current version with the click of a button and export their current models info into the updated model.

When the estimator with an outdated model is loaded, there will be a pop-up notification as shown below.

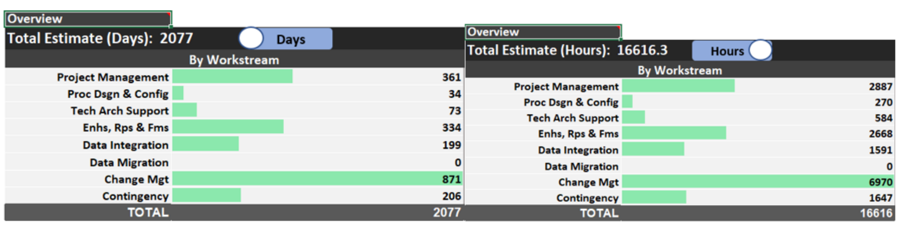


When 'Yes' is selected, a latest version of the model will be downloaded for it to be used.



If the selection is 'No' then only the latest release notes from model version file will be displayed.

2.1.2.12 Toggle Summary between Hours and Days



The toggle feature is present across all the sheet and enables the user to switch the summary of total efforts between days and hours.

2.1.3 Using the Start worksheet

1. Enter the pursuit parameters.
2. Copy all known assumptions to provide a context for estimator review. These will primarily come from the solution plan.
3. Consider distinguishing the original assumptions (color, text or otherwise) from those acquired during the estimating process. This will help to transfer such assumptions into the solution plan once the estimate is complete.
4. Keep track of the assumptions by date they are made.

2.2 Set up and refine overall scope

2.2.1 Overview

In this step, users establish the overall scope to specify the stages of work and workstreams that are in the scope of the estimate, for example, designating whether the effort for the test stage is covered by EY resources (in scope) or by client (out of scope).

The detailed list of stages of work and their definitions can be found in the Digital Delivery Methodology (DDM), and workstreams for a specific solution domain/subdomain can be found in the appendix.

Additionally, users may need to make decisions regarding scope at a lower level of granularity. Examples include whether user acceptance testing (UAT) is in-scope, or the estimate needs to be adjusted because the delivery team resides at three different locations (in which case, the Geographic distribution adjustment should be included in scope).

There are also several estimator parameters that may affect the overall estimate computation and that should be established at the start of the process: amount of contingency, project duration, start date, etc.

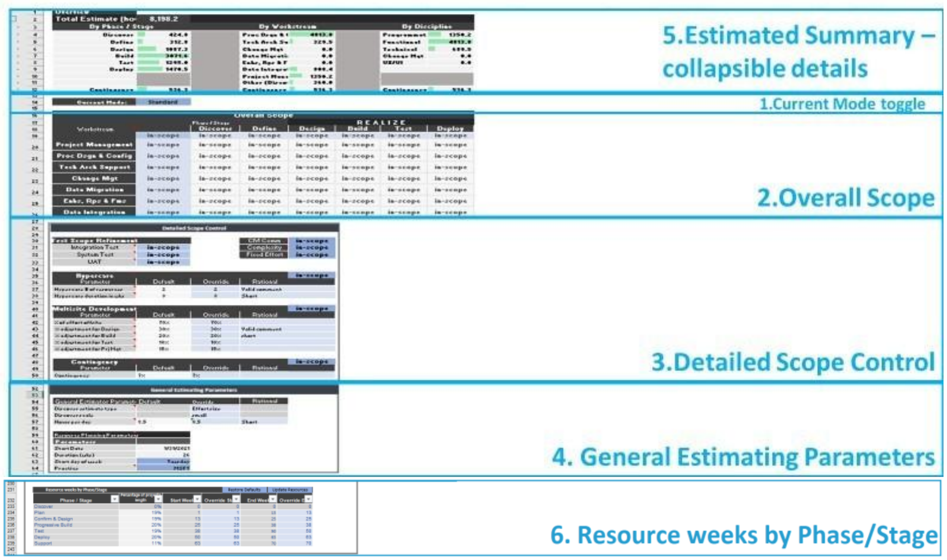
Lastly, if the user needs to change the estimate, there are two modes: standard and basic:

- Standard is the recommended mode for most cases, i.e., during a proposal pursuit.
- Basic, which is designed to remove overhead and several adjustments from the calculation, is for supporting change request estimations. It can also be used as an exploration method for understanding calculation, especially by new users.

The following items will be excluded from estimate calculation in basic mode:

- Project management
- Hypercare
- Change management
- Contingency
- Complexity adjustment
- Geographic distribution adjustment
- Fixed effort calculation (view appendix for more details on fixed effort)

2.2.2 Scope worksheet details



The Scope worksheet emphasizes the overall scope of the effort estimate and establishes or refines key estimating parameters.

- **Current Mode toggle** – toggle to switch between standard and basic mode
- **Overall Scope** – scope estimation for workstreams by stage/phase
- **Detailed Scope Control** – covers test scope refinement as well as adjusting parameters
- **General Estimating Parameters** – mainly involves discover phase and its parameters
- **Estimated Summary – collapsible details** – overall estimation summary
- **Resource weeks by Phase/Stage -** for resourcing purposes, to be used in Resources tab

2.2.2.1 Current Mode toggle

This allows user to switch between basic and standard mode and determines certain workstream and stages to be in scope versus out of scope, eventually impacting the overall estimation.

Current Mc	Basic

	Overall Scope								
	Workstream		Phase / Stage Discover	Plan	Confirm & Design	Progressive Build	Test	Deploy	Support
		in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	out-of-scope
MOBILIZE	Project Management	out-of-scope	out-of-scope	out-of-scope	out-of-scope	out-of-scope	out-of-scope	out-of-scope	out-of-scope
	Business Process Design & Config	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	out-of-scope
	Enhancements	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	out-of-scope
	Reports	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	out-of-scope
	Data Integration	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	out-of-scope
	Data Conversion	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	out-of-scope
	Change Management	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	out-of-scope
	Technical Arcitecture	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	out-of-scope
	Security & Compliance	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	out-of-scope

When basic mode is engaged, project management workstream, discover stage and other parameters become out of scope.

Current Mc	Standard

Overall Scope									
	Workstream		Phase / Stage Discover	Plan	Confirm & Design	Progressive Build	Test	Deploy	Support
		in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
MOBILIZE	Project Management	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Business Process Design & Config	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Enhancements	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Reports	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Data Integration	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Data Conversion	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Change Management	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Technical Arcitecture	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Security & Compliance	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope

When standard mode is engaged, every workstream, stage and parameter become in scope except for Discovery phase.

2.2.2.2 Overall Scope table

Overall Scope									
	Workstream		Phase / Stage Discover	Plan	Confirm & Design	Progressive Build	Test	Deploy	Support
		in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
MOBILIZE	Project Management	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Business Process Design & Config	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Enhancements	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Reports	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Data Integration	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Data Conversion	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Change Management	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Technical Arcitecture	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope
	Security & Compliance	in-scope	out-of-scope	in-scope	in-scope	in-scope	in-scope	in-scope	in-scope

This table is designed to control the scope of estimate by phase/stage and workstream. It is aligned with the phase/stage structure of the Digital Delivery Methodology and the workstream structure of the covered solution domain. Users will be able to select between in scope and out of scope for each workstream and phase individually for more accurate estimation.

A workstream is a basic unit of estimation within the Estimator. Following are some examples of workstreams:

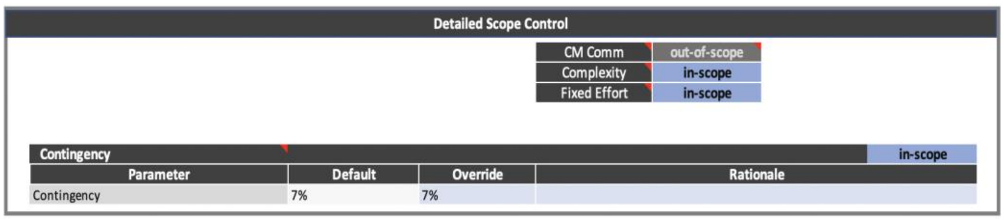
- Project management
- Business Process design and configuration
- Technical architecture
- Change management
- Data migration
- Enhancements
- Reports
- Data conversion

- Security & compliance

Standard phases/stages of work within the Digital Delivery Methodology are listed below:

- Discover
- Plan
- Confirm & design
- Progressive build
- Test
- Deploy
- Support

2.2.2.3 Detailed Scope Control



Detailed Scope Control

CM Comm	out-of-scope
Complexity	in-scope
Fixed Effort	in-scope

Contingency — in-scope

Parameter	Default	Override	Rationale
Contingency	7%	7%	

This section covers test scope refinement for integration testing, system testing and UAT as well as engagement parameters. These individual items can be either in or out of scope, and their default values can also be overridden for estimation:

- **CM comm** — a portion of change management (CM) effort related to communication activities of the initiative that must be supported by project being estimated: Currently out of scope
    - The effort is a small percentage (e.g., 3%) of the overall non-management effort and is added into the project management effort component of the estimate.
- **Complexity** — refers to the adjustment mechanism associated with complexity of the engagement being estimated:
    - This adjustment is performed on workstream basis, with individual complexity drivers contributing to the increase of adjustments (see Complexity worksheet). Switching this mechanism to out of scope allows the estimated effort to be unaffected by complexity settings on the Complexity worksheet.
- **Fixed effort** — sets up effort that varies by workstream and stages of work within the workstreams, e.g., the effort to set up tools and environment; typical on any project but particularly important in a cloud-based environment, such as:
    - Configuring multiple servers, including setting databases: SQL servers, configuring tools (e.g., data factory)
    - Test environment/test data setup
    - Setting stakeholder and end-user interviews (vs. conducting them)
- **Contingency** — allowance made for the risk that something will not be accounted for in the estimate:
    - Contingency parameter depends on the level of risk assumed. It must be switched to in

scope in order to be included in the estimation. When it is in scope, each parameter can be given a default value and override value for the estimation.

2.2.2.4 General Estimating Parameters

General Estimating Parameters

Scalers			in-scope
Workstream/ Scaler	**Default**	**Override/Va**	**Rationale/Comment**
Common Scalers			
Languages	1	1	
Months of Hypercare	1	1	
UAT Cycles	2	2	
SIT Cycles	2	2	
Migration cycles	3	3	
Business Unit	1	1	
Number of Clouds	1	1	
usiness Process Design & Config			
Ledgers	1	1	
Technical Architecture			
Application Landscape	20	20	
Cloud Environments	4	4	
Integration Platforms	1	1	
Security & Compliance			
Legal Entities	1	1	

Assets			in-scope
Asset	**Scope**	**Override/Value**	**Rationale/Comment**
EY EA Data Conversion Recon Scri	in-scope	-10%	
HCM cloud interface tool	in-scope	0%	
RACOR	in-scope	-10%	

Workstream Parameters			
Workstream/ Parameter	**Default**	**Override/Value**	**Rationale/Comment**
Common Parameters			
Oracle Users			
Change Management			
% of training to be delivered via Tra	100%	100%	
% of EY TtT instructors	0%	0%	
Avg Trainees per TtT Class	2.0	2.0	
% of course hours to be translated	0%	0%	
Languages to translate	0.0	0.0	
Average Course Hours per User	24.0	24.0	
Instructors per ILT/VILT class	2.0	2.0	
Support per WBL class	0.1	0.1	
Users per ILT/VILT class	20.0	20.0	
Users per WBL course	150.0	150.0	
% ILT	0%	0%	
% VILT	0%	0%	
% WBL	100%	100%	

Resource Planning Parameters	
Parameter	
Start Date	
Duration (wks)	70
Start day of week	Monday
Practice	Oracle
Hours per day	9.0

Current Curency Selections	
Country Currenc	USA SDC
FX Code	USD
Symbol	$
1 USD =	
1 GDP =	

Resource weeks by Phase/Stage — Restore Defaults | Update Resources

Phase / Stage	Percentage of project's lengt	Start Wee	Override S	End Wee	Override E
Discover	0%	0	0	0	0
Plan	19%	1	1	13	13
Confirm & Design	19%	13	13	25	25
Progressive Build	20%	25	25	38	38
Test	19%	38	38	50	50
Deploy	20%	50	50	63	63
Support	11%	63	63	70	70

These parameters are mainly used for the discover phase for the estimation as well as for resource planning:

- **Scalers** — Types of drivers that amplify impact of specific-size drivers. An increase in unit count of a scalar (e.g., business unit) will increase effort for the impacted driver (e.g., business process) by a specified percentage (e.g., 20%) for each additional unit greater than 1.

  - Common Scalers
    - **Languages** – Number of languages that affect the Business Process Design & Config and Training Performance Support. Overall, increases the base effort for Enhancements, Reports and Data Conversion by 7% for each unit greater than 1. Similarly, increases Business Process and Configuration by 3% for each unit greater than 1. For more information refer to Scaler section in the User Guide.
    - **Months of Hypercare –** The actual duration of hypercare in months, with default being 1 month. Overall, increases the base effort for Business Process Design & Config by 2% for each unit greater than 1. Similarly, increases Security & Compliance and Data Conversion by 3%, Technical Architecture by 4%, Enhancements and Reports by 5%, Data Integration by 6%, and Change Management by 8% for each unit greater than 1. For more information refer to Scaler section in the User Guide.
    - **UAT Cycles –** Increases all RICEFW related workstreams effort in the Test phase of work. Overall, increases the base effort for Enhancements and Reports by 4% for each unit greater than 1. Similarly, increases Data Integration by 5% for each unit greater than 1. For more information refer to Scaler section in the User Guide.
    - **SIT Cycles –** Increases all RICEFW related workstreams effort in the Test phase of work. The effort is increased for each SIT test cycle greater 1. Overall, increases the base effort for Enhancements and Reports by 4% for each unit greater than 1. Similarly, increases Data Integration by 5% for each unit greater than 1. For more information refer to Scaler section in the User Guide.
    - **Migration Cycles -** The number of times a set of conversion program must be re-run, with the goal of reducing number of data conversion errors with each cycle.
    - **Business Units –** Increases the Business Process and Configuration workstream effort for each additional business unit greater than 1 – default value set as 1
    - **Number of Clouds –** Number of Clouds that client is using for their Cloud strategy. For example is client planning to use more than one leading cloud platforms (AWS, GCP, Azure etc.). Excluding Oracle cloud.
  - Business Process Design and Config
    - **Ledgers** – Number of ledgers that affect the Business Process Design & Config. Overall, increases the base effort for Business Process Design & Config by roughly 2% for each unit greater than 1. For more information refer to Scaler section in the User Guide.
  - Technical Architecture
    - **Application Landscape –** Number of different business applications that will be interacting with Oracle Cloud in the future state. For example: Bank, Tax Engine, Credit providers, P-card, Consolidation Engine, EDI engine, Identify Management systems, etc.). Overall, increases the base effort for Technical Architecture by roughly 3% for each unit greater than 5. For more information refer to Scaler section in the User Guide.
    - **Cloud Environments –** Number of Oracle Cloud Environments to be managed during Implementation Cycle. Overall, increases the base effort for Technical Architecture by

roughly 7% for each unit greater than 4. For more information refer to Scaler section in the User Guide.

    - **Integration Platforms –** Number of different Integration Platforms to be considered for Integration Architecture (e.g. MuleSoft, Del Boomi, OIC, etc.). Overall, increases the base effort for Technical Architecture by roughly 11% for each unit greater than 1. For more information refer to Scaler section in the User Guide.

- Security & Compliance

    - **Legal Entities** – The number of Legal Entities that affect Security & Compliance workstream. Typically smaller than the total number of Legal Entities.

- **Assets** –

    - **EY EA Data Conversion Recon Scripts –** Recon scripts for Oracle Cloud rapid implementations assist business process team to validate configurations/setups rather than screen compare. Reduces Test effort by 10% in the Business Process Design & Config workstream.

    Usage Criteria:

    1. No installation required, BIP reports need to be migrated to target instance to execute and generate reconciliation report

    2. This should be preferred when RACOR is NOT used

    - **HCM cloud interface tool –** Solution for Oracle HCM Cloud implementations using the source system's data and migrate the same to Oracle HCM Cloud.

    Efforts need to be manually factored under Data Conversion Workstream or HCM module effort

    Usage Criteria: Selecting this asset would not automatically reduce the efforts on the estimator instead manual intervention is required

    - **RACOR** - A single spot solution for Oracle Cloud rapid implementations using the source system's data and migrate the same to Oracle Cloud.

- **Workstream Parameters** — these work in conjunction with the size drivers and help estimate the base effort of the individual workstreams

    - Common Parameters

        - **Oracle Users** – Total number of Oracle system users

    - Change Experience Parameters

        - **% of training to be delivered via Train-the-Trainer method** – Percentage of training to be delivered via Train-the-Trainer method. The intent is to deliver as much training is possible using this method with most instructors to be provided by the client - see % of EY TtT instructors parameter

        - **% of EY TtT instructors** – Percentage of TtT instructors provided by EY. The intent of to have most of the instructors to be provided by client, hence the default is 0%.

        - **Avg Trainees per TtT Class** – Average number of Trainers to be trained via TtT method. Must be at least 2 trainees per class - which leads to a 1-on-1 ratio for instructors to trainees

        - **% of course hours to be translated** – % of training to be translated. This applies to both training courseware material and performance support material.

- **Languages to translate** – Number of target languages to translate the training
- **Average Course Hours per User** – Average amount of training course hours to be delivered to the individual users. The amount to be delivered via different methods is determined by %ILT, % VILT, %WBL listed below.
- **Instructors per ILT/VILT class** – Expected number of instructors required to lead ILT or VILT training
- **Support per WBL class** – An average amount of effort (fraction of an hour) needed to support a WBL run.
- **Users per ILT/VILT class** – Average number of users per ILT or VILT class.
- **Users per WBL course** – Average number of users that will take a single WBL course
- **% ILT** – % of expected training to be ILT
- **% VILT** – % of expected training to be VILT
- **% WBL** – % of expected training to be WBL

▸ **Resource planning parameters** — The following parameters govern the display and management of time-dependent items on the Resources worksheet:

- ▸ **Start date** — the starting date for the resources at the engagement
- ▸ **Duration (weeks)** — number of weeks the resources are planned for the engagement
- ▸ **Start day of week** — the actual day when the resources will start in the first week
- ▸ **Practice** — name of the practice the engagement belongs to
- ▸ **Hours per day** — the average working hours per day

▸ **Current Currency Selections**

- ▸ **Country Currency** – country of currency
- ▸ **FX Code** – foreign exchange global code for that country
- ▸ **Symbol** – symbol of the currency
- ▸ **1 USD =** - how much is 1 USD equivalent to the currency USD is being converted into
- ▸ **1 GDP =** - how much is 1 GDP of that country equivalent to USD

2.2.2.5 Estimate Summary — collapsible details

Total Estimate (hours): 615.4

By Phase / Stage		By Workstream		By Discipline	
Discover	0	Project Management	575	Programmatic	575
Plan	108	Business Process Design & Config	0	Functional	0
Confirm & Design	0	Enhancements	0	Technical	0
Progressive Build	208	Reports	0	Change Mgt	0
Test	182	Data Integration	0	UX/UI	0
Deploy	65	Data Conversion	0		
Support	13	Change Management	0		
		Technical Arcitecture	0		
		Security & Compliance	0		
Contingency	40	Contingency	40	Contingency	40
TOTAL	615		615		615

The total effort estimate is always displayed at the top of the worksheet. However, the details normally hidden can be expanded by clicking on a "+" sign. Once expanded, this section contains the estimate summary broken down by phases/stages, workstreams and disciplines. The detailed description of these is identical for all worksheets and can be found under *Estimate*.

2.2.2.6 Resource weeks by Phase/Stage

Resource weeks by Phase/Stage		Restore Defaults	Update Resources		
Phase / Stage	Percantage of project's length	Start Week	Override Start	End Week	Override End
Discover	0%	0	0	0	0
Plan	19%	1	1	13	13
Confirm & Design	19%	13	13	25	25
Progressive Build	20%	25	25	38	38
Test	19%	38	38	50	50
Deploy	20%	50	50	63	63
Support	11%	63	63	70	70

This Assign Phases for Weeks button allows user to schedule the Phases/Stages of the project within each week while preparing resources for it. User is able to modify Start Week and End Week of each Phase. There is a data validation within the entry fields that enables user to submit only positive numbers, where Start Week is lower or equal to End Week of current Phase. After changing values, user can always revert to the default one by clicking a 'Default' button on the right-hand side from the field.

- **Restore Defaults** - changes all values of the Start and End Weeks to the default ones
- **Update Resources** - saves latest values and displays Phases two rows above each week in **Resources Canvas** based on the schedule. If there are more than 2 Phases within one week there will be displayed only the first one with continuation sign.

Important: this same functionality is also included in Resourcing canvas under "Assign Phases for Weeks" button. Please refer to point 3.2.2.5

2.2.3 Using the Scope worksheet

Proceed with the following steps to complete this worksheet:

1. **Overall scope** — Select stages and workstreams that are in scope in the user's estimate.
2. **Testing scope** — Typically, all test types are included in scope. Exclude the specific test types if user knows client will be responsible for their execution.
3. **Change management communication** — Decide on whether to include this item in scope based on the nature of the proposal/project and client's need for a broad-based communication tied to the umbrella initiative.
4. **Complexity** — In a typical scenario, consider leaving this in scope. Excluding it from scope will disable the adjustments associated with engagement complexity (Complexity worksheet) for all workstreams.
5. **Fixed effort** — Keep this item in scope for most typical scenarios. The only exception is when user considers combining multiple separately created estimates and would like to avoid double counting this effort.
6. **Hypercare** — Determine whether to keep this item in scope based on the client's requirements. In most scenarios, user may want to keep it in scope and adjust the parameters in the hypercare section. By default, it assumes two resources for eight weeks.
7. **Geographic Distribution parameters** — Include this in scope if user expects the development team to reside in multiple locations. First, consider the amount of effort being completed at other locations and set percentage of effort off-site accordingly. Then, evaluate the default parameters for each stage of work and fine-tune them based on the user's project/proposal.

Change the adjustors if this a long-running engagement and user has experience with the team and feels comfortable making changes. Otherwise, leave the default settings in place.

8. **Contingency** — There are two major ways of using contingency: (1) to cover unknowns and (2) a formal approach for using it as a solution contingency to resolve potential risks. Choose it based on the specifics of user proposal/project.
9. **General estimating parameters** — Intended to be self-explanatory. It is recommended to walk through and set each parameter at the start of estimation. Two parameters require more explanation: discover estimate type and discover scale. First, choose how user would like to estimate the discover phase: as small, medium or large fixed effort (by size) or as a percentage of the rest of the effort, excluding project management. If effort size is chosen, then select small, medium or large in the discover phase parameter. If percentage of rest stages is chosen, then it is automatically calculated.

2.3 Set up and refine functional scope

2.3.1 Overview

This step enables the user to align the scope of the estimate with the functional scope of the proposal/project. In the context of estimating, the functional scope is determined by the number of business processes included in scope, and that number will drive the process design and configuration effort.

Set up and refine the functional scope of the estimate by selecting portions of the business process hierarchy that is to be included in the scope of the estimate. There are three levels in the business process hierarchy: L1 — top-level process, L2 — middle-tier, and L3 — lower-level process. The level business process drives the effort computation.

Users can change the size setting of the individual business process from its default setting. Users can also make adjustments at the aggregate level for all L3 business processes of given L1 business process, but users cannot do this at the individual business process basis. They can, for example, reduce the effort of all L3 business processes within procure-to-pay by some percentage.

Review entire hierarchy and decide on which L1, L2 and L3 processes should be estimated and toggle them in and out of scope. Portions of the process hierarchy can be removed from scope by toggling the scope settings of the individual L2 or L3 business processes.

Users can review the total estimated hours associated with the individual L1 business processes.

Users can also edit the process hierarchy and add entirely new L1 processes, add/remove L2 process branches to/from an existing L1 business process or add/remove L3 business processes underneath the existing L2 business process.

In addition, users can establish linkages between L1 business processes and the other components of user's estimate — refer to detail steps under the Component worksheet – section *2.4.2.1*. Once a user has established the linkage, toggling an L1 business process in or out of scope will automatically toggle all linked components.

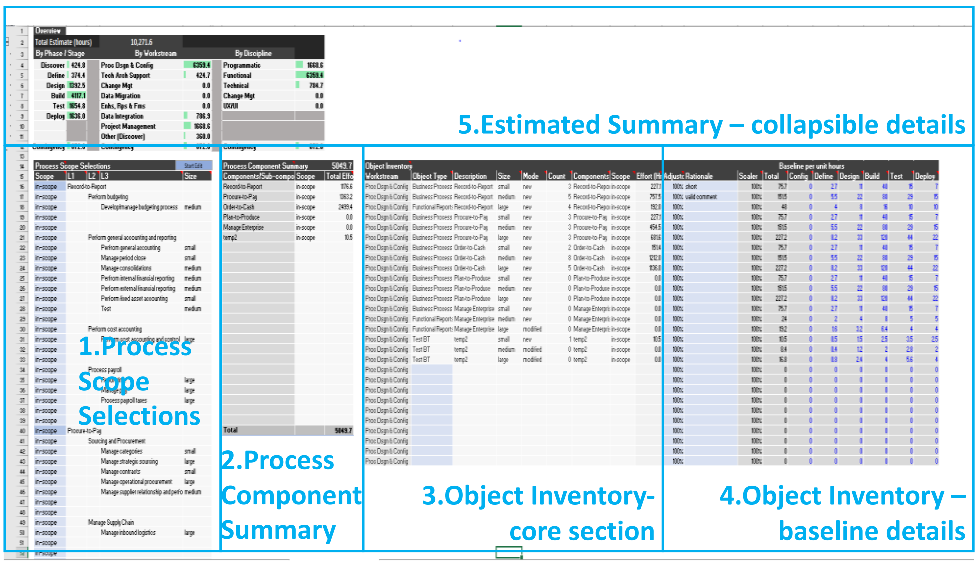
The following rules apply for including/excluding business process in or out of scope:

- L1 processes can be included/excluded from scope with entire portion of the process hierarchy being excluded/include into/from scope.
- L2 and L3 business processes can be excluded from the scope of the estimate if the parent L1 process is included in scope.
- Converse is **not** true: L2 and L3 processes cannot be included into the scope of estimate if the parent L1 process is not in scope. Whether they are in scope or out of scope or their complexity is

small, medium or large will impact the total effort estimate, which is clearly visible under Process Component Summary table.

To execute this step, navigate to the Functional worksheet.

2.3.2 Functional worksheet Details



This worksheet is broken down into the following sections:

- **Process Scope Selections** – based on a hierarchy where estimation relies on the functional scope of L1 business process; could be done in a more in-depth manner where both L2 and L3 processes are involved and either of their scope differs from L1 process
- **Process Component Summary** – provides a list of components or L1 business processes with their corresponding scope status and effort hours
- **Object Inventory – core section** – enables entry of an inventory of objects (size drivers) that ultimately drive effort for the project; this portion of the table is always visible
- **Object Inventory – baseline details** – allows viewing of additional detail
- **Estimate Summary – collapsible details** – overall estimation summary

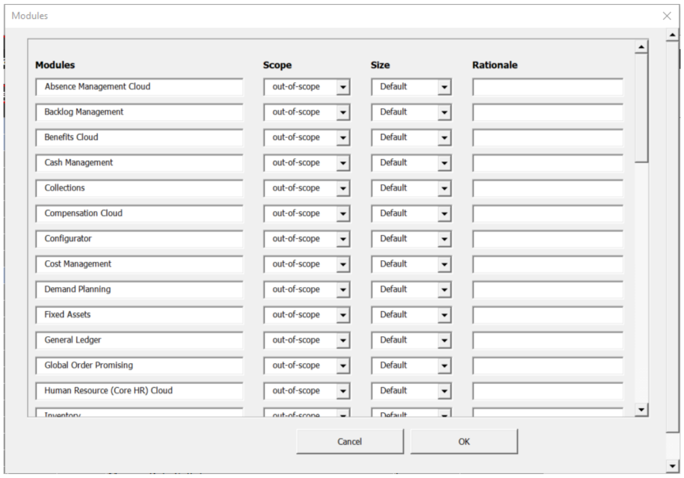
2.3.2.1 Process Scope Selections

	Process Scope Selections			Estimate Pivot Table / Start Edit		Modules	
Module	**Scope**	**L1**	**L2**	**L3**	**Size**	**Adjustor**	**Rationale**
	in-scope	RICEFW Functional Effort				100.00%	
Absence Manage	in-scope	Application Platform and Infrastructure				100.00%	
	in-scope		Process Management and Integration			100.00%	
	in-scope			Configure Absence	small	100.00%	
	in-scope			Maintain Absence Enrollments	small	100.00%	
	in-scope			Manage Absence Records	small	100.00%	
	in-scope			Manage Absence Cases	small	100.00%	
	in-scope			Manage Absence Processing	small	100.00%	
	in-scope			Report and Analyze Absence Data	small	100.00%	
Backlog Manage	in-scope	Supply Chain Planning (BM)				100.00%	
	out-of-scope		Manage Backlog Planning			100.00%	
	out-of-scope			Maintain Backlog Planning Process	large	100.00%	
	out-of-scope			Execute Backlog Planning	large	100.00%	
	in-scope		Manage Supply Allocation rules			100.00%	
	in-scope			Define a supply allocation rule	medium	100.00%	
	in-scope		Sourcing rules, Bills of distribution & Assignment Sets			100.00%	
	out-of-scope			Manage Sourcing rules & Bills of distribution	large	100.00%	
	in-scope			Manage assignment set	medium	100.00%	

- **Start edit/save changes**:
  - *Start Edit* enables editing of the business process hierarchy.
  - *Save Changes* finalizes the user's edits and updates the workbook.
- **Scope** — enables business users to change the L1, L2 and L3 processes' scope status:
  - By default, L2 and L3 process will follow scope selection of L1 business process. However, individual L2 and L3 processes scope selection can be changed to out of scope only if L1 process scope is in scope but cannot be done in reverse manner.
- **Estimate Pivot Table** – this shows a new worksheet with pivot table. The detailed description can be found in its own chapter 2.6
- **L1** — the highest (top-level nodes) business processes in the hierarchy
- **L2** — the mid-level nodes in the business processes hierarchy
- **L3** — the lowest nodes in the business process hierarchy; only level of business processes that drives count, and consequently effort, for business process design and configuration
- **Size** — determines the size of the corresponding L3 business process (small, medium and large)
- **Adjustor** – allows user to adjust individual L3 business process and their effort
- **Rationale** – User can provide explanation based on their changes

Modules –

Each L1 in the business process is scoped to the module listed in the column to its left. The Modules button opens a user form that allows users to change the scope and size selections for all processes associated with a given module and provide a rationale for why the updates were made.



- **Modules** – list of the available modules
- **Scope** – scope selection for all processes associated with the module
- **Size** – selection that determines the size for all L3 business processes associated with the module
    - When small, medium, or large selection is chosen then all L3 business processes associated with the module are set to the specified size
    - When the default selection is chosen, then all L3 business processes associated with the module are returned to their original values
- **Rationale** – textbox to provide an explanation for the selections chosen for the module

2.3.2.2 Process Component Summary

Process Component Summary		2	7	445
Components/Sub components	**Scope**	**L2 Count**	**L3 Count**	**Total Effort**
RICEFW Functional Effort	out-of-scope	0	0	0
Application Platform and Infrastructure	in-scope	1	6	386
Supply Chain Planning (BM)	in-scope	1	1	59
Compensation Management (BC)	out-of-scope	0	0	0
Cash and Treasury Management	out-of-scope	0	0	0
Order Fulfilment (CoI)	out-of-scope	0	0	0
Compensation Management (CC)	out-of-scope	0	0	0
Order Fulfilment (Con)	out-of-scope	0	0	0
Materials Management and Logistic (CM)	out-of-scope	0	0	0
Order Fulfilment (CM)	out-of-scope	0	0	0
Product Management (CM)	out-of-scope	0	0	0
Production (CM)	out-of-scope	0	0	0
Supply Chain Planning (DP)	out-of-scope	0	0	0
Asset Lifecycle Management	out-of-scope	0	0	0
Maintain Enterprise Structure	out-of-scope	0	0	0
Maintain 4 C's	out-of-scope	0	0	0

- **Components/subcomponents** – name of components or subcomponents that align with L1 business processes from the Process Scope Selections table
- **Scope** – scope status for the list of components:
  - If the scope status for a component is changed in the Process Scope Selections table, it would automatically impact the same component in this Process Component Summary table.
- **L2 Count** - number of L2 count for each L1 business process and total sum
- **L3 Count** - number of L3 count for each L1 business process and total sum
- **Total effort** – estimated total effort for individual L1 processes in hours and the cumulative sum on top of the column

2.3.2.3 Object Inventory – core section

Object Inventory									
Workstream	**Sub Workstream**	**Object Type**	**Description**	**Size**	**Mode**	**Count**	**Components**	**Scope**	**Effort (Hrs)**
Business Process Design & Config	Business Process Design & Config	Business Process	Manage Trans	small	new	2	Manage Transactiona	in-scope	50.5
Business Process Design & Config	Business Process Design & Config	Business Process	Manage Trans	medium	new	3	Manage Transactiona	in-scope	177.0
Business Process Design & Config	Business Process Design & Config	Business Process	Manage Trans	large	new	1	Manage Transactiona	in-scope	92.0
Business Process Design & Config	Business Process Design & Config	Business Process	Manage Suble	small	new	0	Manage Subledger Ac	in-scope	0.0
Business Process Design & Config	Business Process Design & Config	Business Process	Manage Suble	medium	new	9	Manage Subledger Ac	in-scope	531.0
Business Process Design & Config	Business Process Design & Config	Business Process	Manage Suble	large	new	1	Manage Subledger Ac	in-scope	92.0
Business Process Design & Config	Business Process Design & Config	Business Process	Manage Sour	small	new	3	Manage Source Syster	in-scope	75.7
Business Process Design & Config	Business Process Design & Config	Business Process	Manage Sour	medium	new	2	Manage Source Syster	in-scope	118.0
Business Process Design & Config	Business Process Design & Config	Business Process	Manage Sour	large	new	3	Manage Source Syster	in-scope	276.0
Business Process Design & Config	Business Process Design & Config	Business Process	Workforce De	small	new	1	Workforce Deployme	in-scope	25.2
Business Process Design & Config	Business Process Design & Config	Business Process	Workforce De	medium	new	2	Workforce Deployme	in-scope	118.0
Business Process Design & Config	Business Process Design & Config	Business Process	Workforce De	large	new	1	Workforce Deployme	in-scope	92.0
Business Process Design & Config	Business Process Design & Config	Business Process	Workforce De	small	new	5	Workforce Developm	in-scope	126.1
Business Process Design & Config	Business Process Design & Config	Business Process	Workforce De	medium	new	8	Workforce Developm	in-scope	472.0
Business Process Design & Config	Business Process Design & Config	Business Process	Workforce De	large	new	3	Workforce Developm	in-scope	276.0
Business Process Design & Config	Business Process Design & Config	Business Process	Supply Chain	small	new	0	Supply Chain Plannin	in-scope	0.0

- **Workstream** – a basic unit of estimation within the estimating tool. Select the name of a workstream from a drop-down list.
- **Object Type** – specific to the work stream for a given line item. Select the desired object type from the drop-down list
- **Description** – an optional field used to describe the object. Select type in the description.
- **Size** – differentiates the effort for an object type for a given line item – *small*, *medium* or *large*. Select size from the drop-down list.
- **Mode** – determines if a given line item needs to be built from scratch and can be adapted based on an existing artifact (*new* or *modified*). Select mode from the drop-down list.
- **Count** – the number of objects selected in each line item. Enter a number into the field.
- **Component** – a name of the desired component with which a given object is associated. Select the name of a component from a drop-down list.
- **Scope** – indicates whether a given object is in scope of the estimate. This is a read-only field that depends on the status of the parent component. Produces 0 estimate when out of scope.
- **Effort (hours)** – total effort associated for each object. This is a computed field.

2.3.2.4 Object Inventory – baseline details

			Baseline per unit hours						
Adjustor	**Rationale**	**Total**	**Discover**	**Plan**	**Confirm & D**	**Progressive**	**Test**	**Deploy**	**Support**
100%		25.23	0.00	2.42	5.00	8.00	6.50	2.80	0.50
100%		59.00	0.00	5.50	13.00	17.00	15.50	6.50	1.50
100%		92.00	0.00	9.00	21.00	26.00	24.00	10.00	2.00
100%		25.23	0.00	2.42	5.00	8.00	6.50	2.80	0.50
100%		59.00	0.00	5.50	13.00	17.00	15.50	6.50	1.50
100%		92.00	0.00	9.00	21.00	26.00	24.00	10.00	2.00
100%		25.23	0.00	2.42	5.00	8.00	6.50	2.80	0.50
100%		59.00	0.00	5.50	13.00	17.00	15.50	6.50	1.50
100%		92.00	0.00	9.00	21.00	26.00	24.00	10.00	2.00
100%		25.23	0.00	2.42	5.00	8.00	6.50	2.80	0.50
100%		59.00	0.00	5.50	13.00	17.00	15.50	6.50	1.50
100%		92.00	0.00	9.00	21.00	26.00	24.00	10.00	2.00
100%		25.23	0.00	2.42	5.00	8.00	6.50	2.80	0.50
100%		59.00	0.00	5.50	13.00	17.00	15.50	6.50	1.50
100%		92.00	0.00	9.00	21.00	26.00	24.00	10.00	2.00
100%		25.23	0.00	2.42	5.00	8.00	6.50	2.80	0.50

- **Adjustor** – allows adjustment of the corresponding line item:
  - Change percentage and then add rationale in the Rationale column.
- **Rationale** – provides rationale for having an adjustor value different than 100%:
  - If no rationale is entered, the entry is highlighted red. Otherwise, the adjustor and rationale entries are highlighted in yellow, capturing the attention of a reviewer.
- **Scaler** – adjusts the estimation via a scaler factor; currently not used
- **Total** – the total effort for a single unit associated with a given line item
- **Define** – hour-per-unit value associated with the effort in define stage for the specified size of a given line item

- **Design** — hour-per-unit value associated with the effort in design stage for the specified size of a given line item
- **Build** — hour-per-unit value associated with the effort in build stage for the specified size of a given line item
- **Test** — hour-per-unit value associated with the effort in testing stage for the specified size of a given line item
- **Deploy** — hour-per-unit value associated with the effort in deploy stage for the specified size of a given line item

2.3.2.5 Estimate Summary — collapsible details

Total Estimate (hours): 615.4					
By Phase / Stage		By Workstream		By Discipline	
Discover	0	Project Management	575	Programmatic	575
Plan	108	Business Process Design & Config	0	Functional	0
Confirm & Design	0	Enhancements	0	Technical	0
Progressive Build	208	Reports	0	Change Mgt	0
Test	182	Data Integration	0	UX/UI	0
Deploy	65	Data Conversion	0		
Support	13	Change Management	0		
		Technical Arcitecture	0		
		Security & Compliance	0		
Contingency	40	Contingency	40	Contingency	40
TOTAL	615		615		615

The total effort estimate is always displayed at the top of the worksheet. However, the details normally hidden can be expanded by clicking on a “+” sign. Once expanded, this section contains estimate summary broken down by phases/stages, workstreams and disciplines. The detailed description of these is identical for all worksheets and can be found under *Estimate*.

2.3.3 Using the Functional worksheet

Steps for completing this worksheet:

1. Walk through the Scope column in the Process Scope Selection table, changing in- and out-of-scope settings of the individual L1 business processes.
2. Fine-tune individual L1 business processes by toggling scope settings of the individual L2 and L3 business process.
3. If needed, adjust the size settings of the individual L3 business processes.
4. Review total estimate hours for the individual L1 business processes by walking through the Total Effort column in the Process Component Summary table.
5. Review the total business process design and configuration effort in the summary section — expand the section to view the details for the workstream.
6. If there are missing business processes, add them to the hierarchy. Click on “Start Edit” and make edits as necessary, then click “Save” when done.
7. In the cases of significant over or underestimation for specific business processes, adjust by expanding the section on the right and make changes to the Adjustor column — provide the rationale for user change. Note that this will change simple, medium or large business process effort in the given L1 process. Collapse the section when done.

2.4 Enter and adjust size drivers

2.4.1 Overview

Entering size drivers involves two discrete activities:

- Setting up a component structure
- Entering the inventory of size drivers

Size drivers or physical objects are base elements that drive the bulk of the estimate. In this document, size drivers and objects will be used interchangeably. The notion of objects is related to the fact that the total effort estimate depends on what users need to build, e.g., physical objects or project artifacts.

Each row in the Object table is referred to as a "line item." A line item can represent a collection of objects with the same characteristics. It enables creating an initial estimate quickly, grouping specific object types in categories by size or by mode (see definitions below).

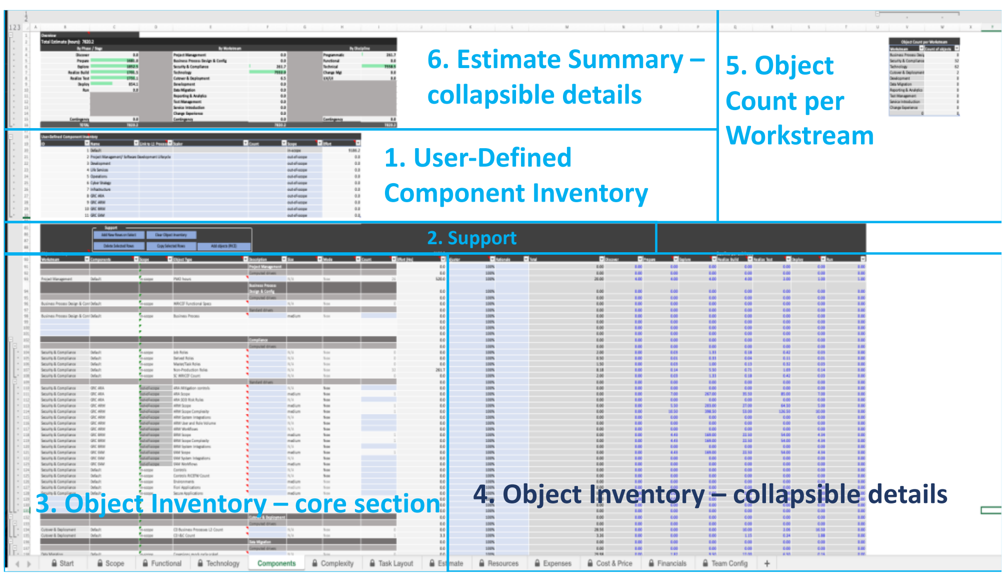
A component is an organizing element enabling a user to group a set of objects into logical collection and then handling such collection or sets of objects efficiently as a group without having to deal with them individually. For example, instead of removing individual objects in or out of scope based on client request, users can simply disable a component, and the effort associated with all related objects will be removed from the estimate.

A component can correspond to a collection of similar objects such as all reports, all outbound interfaces or an entire set of training materials. Alternatively, a component can represent a functional group of objects, including all development objects built to support a functional area.

When estimating small projects, creating separate components may not be necessary. For such simple cases, there is a default component, with which the entire object inventory can be associated – use this component as a reference in such a case.

The Components worksheet is intended to help users assemble and manage an inventory of objects and components. Objects and components are entered into two similarly named tables described below.

2.4.2 Components' worksheet details



The Component worksheet assembles an inventory of objects and components within a workstream. The following sections emphasize major features of this worksheet:

- **User-Defined Component Inventory** — enables entry of an inventory of components that represent groups of objects that are part of the user's estimate
- **Support** — this section allows user to perform multiple functionalities such as adding new rows, deleting rows, copy and paste rows as well as clearing entire table
- **Object Inventory — core section** — enables entry of an inventory of objects (size drivers) that ultimately drive effort for the project; this portion of the table is always visible
- **Object Inventory — collapsible details** — this section is normally collapsed but can be expanded to enable viewing of additional details
- **Object Count per Workstream** — this table displays names of workstreams and their object count
- **Estimated Summary — collapsible details** — overall effort estimation summary

2.4.2.1 User-Defined Component Inventory

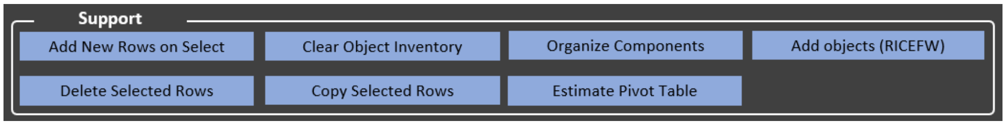
User-Defined Component Inventory

ID	Name	Link to L1 Process	Scaler	Count	Scope	Effort
1	Default				in-scope	3446.0
2	Org Design				in-scope	0.0
3	Training				in-scope	0.0
4	Performance Support				in-scope	0.0
5	Communication				out-of-scope	0.0
6	Project Management/ Software Development Lifecycle				out-of-scope	0.0
7	Development				out-of-scope	0.0
8	Production				out-of-scope	0.0
9	Operations				out-of-scope	0.0

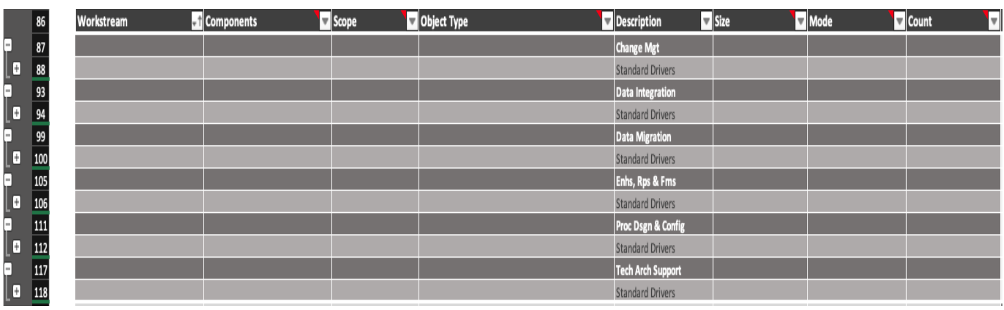
- **ID** — a simple numeric index for ease of identification and review; populated automatically
- **Name** — title of the component that easily identifies a group of objects; typed in by user
- **Link to L1 process** — allows for tight linking of a component to an L1 business process, such as procure-to-pay:
  - User can select a parent L1 process from a drop-down. To remove linkage, simply delete the content of the cell.
- **Scaler** — name of each scaler connected to the specific component
- **Count** — given count for each component and its scaler
- **Scope** — indicates whether the component is in or out of scope:
  - User can manually change the indicator by selecting an appropriate state from a drop-down. Once user links the corresponding component to an L1 process, the indicator will follow the state of the linked-to process. User can partially control the scope of a linked component and can take it out of scope if the parent L1 process is in scope, but user cannot bring a linked out-of-scope component back into scope.
- **Effort —** displays the total estimated effort of the component, which is a sum of the effort for all objects that are associated with such a component, read-only field

  This section might be collapsed, and users can click on the (+) sign on the left side of the worksheet to expand this table.

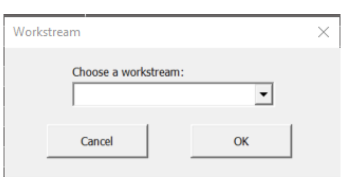
2.4.2.2 Support

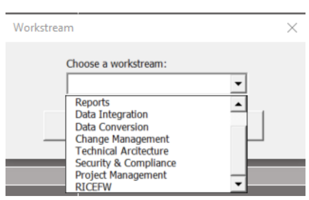


- **Add New Rows on Select** – allows users to add one or multiple rows upon deciding where the newly added rows should be added
- **Delete Selected Rows** – upon selecting one or multiple rows, this button allows user to delete these rows from the table
- **Clear Object Inventory** – this button allows user to clear out the entire object inventory table
- **Copy Selected Rows** – once user selects one or more rows/cells and click on this button, they can select which rows they would like to copy and then paste those rows in the table underneath the Workstream column
- **Organize Components** – this particular feature allows user to organize object inventory table by workstreams and their added size drivers. Once user clicks this button, all added objects will be collapsed within their workstream and can only be seen by expanding on the left side of the worksheet as the picture shows below:

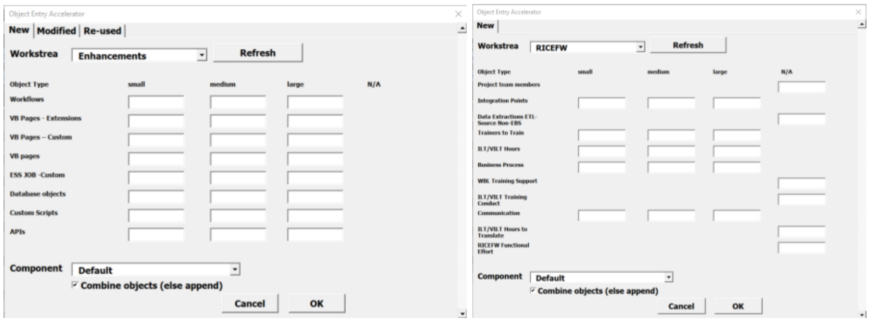


- **Estimate Pivot Table** – this shows a new worksheet with pivot table. The detailed description can be found in its own chapter 2.6
- **Add Objects (RICEFW)** – allows users to quickly add multiple objects to the Object Inventory – core section





1. Select a workstream or RICEFW from the dropdown on the first popup
2. Click OK to open the 'Object Entry Accelerator' window

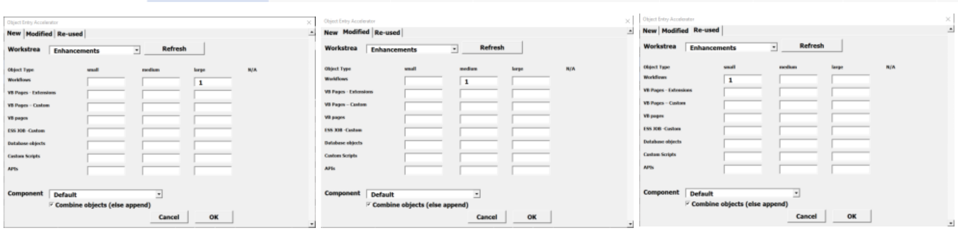


- **New, Modified and Re-use tabs** - The tabs on the top of the window correspond to the Mode selection of the objects. Some objects can only have New as their Mode.
- **Workstream Dropdown** – List of workstreams and RICEFW options
- **Refresh** – After selecting a different option from the Workstream Dropdown, click Refresh to load in the objects associated with the new selection
- **Object Type** – List of objects specific to the chosen workstream/RICEFW objects
- **small, medium, large and N/A** – These headers correspond the Size selection of the object. Users enter object counts into their corresponding textbox under the chosen Size header. Some objects have no Size, so users enter object counts into the corresponding textbox under the N/A header.
- **Component Dropdown** – List of Component names with which the objects can be associated with. To have more selections in the dropdown, add more Components to the User-Defined Component Inventory.
- **Combine objects (else append) Checkbox** – Lets users choose how objects are added to the Object Inventory Table
  - **When Checked** – Objects will be added in bulk with one line item and an aggregated count for each unique Mode, Size combo.

				Reports - 1			
				Additional Drivers			
Reports	Default	in-scope	BI reports		small	new	5

  - **When Unchecked** – Objects will be added line by line with a count of 1.

				Reports - 2			
				Additional Drivers			
Reports	Default	in-scope	BI reports		small	new	1
Reports	Default	in-scope	BI reports		small	new	1
Reports	Default	in-scope	BI reports		small	new	1
Reports	Default	in-scope	BI reports		small	new	1
Reports	Default	in-scope	BI reports		small	new	1



- To add 1 New, large Workflows object, users would enter 1 into the first box under the large header. To add 1 Modified, medium Workflows object, the user would click on the

Modified tab and then enter 1 into the first box under the medium header. Similarly, to add 1 Re-used, small Workflows object, the user would click on the Re-used tab and then enter 1 into the first box under the small header. Click OK to see the three objects added to the Object Inventory table:

Object Inventory Workstream	Components	Scope	Object Type	Description	Size	Mode	Count
				Project Management			
				Standard drivers			
				Enhancements			
				Standard drivers			
				Reports			
				Standard drivers			
				Data Integration			
				Standard drivers			
				Data Conversion			
				Standard drivers			
				Change Management			
				Computed drivers			
				Standard drivers			
				Technical Arcitecture			
				Services			
				Support & Integration			
				Security & Compliance			
				Computed drivers			
				Enhancements - 1			
				Additional Drivers			
Enhancements	Default	in-scope	Workflows		large	new	1
Enhancements	Default	in-scope	Workflows		medium	modified	1
Enhancements	Default	in-scope	Workflows		small	re-used	1

2.4.2.3 Object Inventory – core section

Object Inventory Workstream	Components	Scope	Object Type	Description	Size	Mode	Count	Effort (Hrs)
				Project Management				0.0
				Standard drivers				0.0
Project Management	Default	in-scope	Communication hours		N/A	new	26	575.1
								0.0
				Enhancements				0.0
				Standard drivers				0.0
Enhancements	Default	in-scope	APIs		medium	new	3	673.5
Enhancements	Default	in-scope	Database objects		medium	new	4	860.0
Enhancements	Default	in-scope	VB pages		medium	new	3	936.0
Enhancements	Default	in-scope	Workflows		medium	new	2	393.0
Enhancements	Default	in-scope	VB Pages – Custom		medium	new	4	816.0
Enhancements	Default	in-scope	VB Pages - Extensions		medium	new	4	730.0
Enhancements	Default	in-scope	ESS JOB -Custom		medium	new	3	82.3
Enhancements	Default	in-scope	Custom Scripts		medium	new	3	162.4

- **Workstream** – name of the basic unit of estimation within the Estimator. Select the name of a workstream from a drop-down list.
- **Component** – a name of the desired component with which a given object is associated with. Select the name of a component from a drop-down list.
- **Scope** – indicates whether a given object is in scope of the estimate. This is a read-only field that depends on the status of the parent component. Produces 0 estimate when out of scope.
- **Object type** – specific to the work stream for a given line item. Select the desired object type from the drop-down list.
- **Description** – an optional field that can be used to describes the object. Select type in the description.
- **Size** – differentiates the effort for an object type for a given line item. Select *small*, *medium* or *large* from the drop-down list.
- **Mode** – determines if a given line item needs to be built from scratch and can be adapted based on an existing artifact (*new* or *modified*). Select mode from the drop-down list.
- **Count** – the number of objects selected in each line item. Enter a number into the field.
- **Effort (hours)** – total effort associated for each object. This is a computed field.

2.4.2.4 Object Inventory — collapsible details

					Baseline per unit hours				
Adjustor	Rationale	Total	Discover	Plan	Confirm & Design	Progressive Build	Test	Deploy	Support
100%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100%		22.12	0.00	4.14	0.00	8.00	7.00	2.48	0.50
100%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100%		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100%		224.50	0.00	17.00	45.00	74.00	48.50	27.50	12.50
100%		215.00	0.00	16.00	40.00	73.50	46.50	26.50	12.50

- **Adjustor** — allows adjustment of the corresponding line item:
  - Change percentage and then add explanation in the Rationale column.
- **Rationale —** provides reason for having an adjustor value different than 100%:
  - If no rationale is entered, the entry is highlighted red. Otherwise, the Adjustor and Rationale entries are highlighted yellow, capturing the attention of a reviewer.
- **Scaler** — adjusts the estimation via a scaler factor; currently not used
- **Total** — the total effort for a single unit associated with a given line item
- **Config —** hour-per-unit value associated with the effort in configuration stage for the specified size of a given line item
- **Define —** hour-per-unit value associated with the effort in define stage for the specified size of a given line item
- **Design** — hour-per-unit value associated with the effort in design stage for the specified size of a given line item
- **Build —** hour-per-unit value associated with the effort in build stage for the specified size of a given line item
- **Test** — hour-per-unit value associated with the effort in testing stage for the specified size of a given line item
- **Deploy —** hour-per-unit value associated with the effort in deploy stage for the specified size of a given line item

2.4.2.5 Object Count per Workstream

Object Count per Workstream	
Workstream	Count of objects
Business Process Design	141
Enhancements	26
Reports	0
Data Integration	0
Data Conversion	0
Change Management	4
Technical Arcitecture	0
Security & Compliance	2453
0	0

- **Workstream** — name of workstreams in the object inventory table
- **Count of objects —** number of objects per workstreams

2.4.2.6 Estimate Summary — collapsible details

Total Estimate (hours): 615.4					
By Phase / Stage		By Workstream		By Discipline	
Discover	0	Project Management	575	Programmatic	575
Plan	108	Business Process Design & Config	0	Functional	0
Confirm & Design	0	Enhancements	0	Technical	0
Progressive Build	208	Reports	0	Change Mgt	0
Test	182	Data Integration	0	UX/UI	0
Deploy	65	Data Conversion	0		
Support	13	Change Management	0		
		Technical Arcitecture	0		
		Security & Compliance	0		
Contingency	40	Contingency	40	Contingency	40
TOTAL	615		615		615

The total effort estimate is always displayed at the top of the worksheet. However, the details normally hidden can be expanded by clicking on a "+" sign. Once expanded, this section contains estimate summary broken down by phases/stages, workstreams and disciplines. The detailed description of these is identical for all worksheets and can be found under *Estimate*.

2.4.3 Using the Components worksheet

There are four different ways to proceed:

1. **Top down**, by creating a list of components first and then creating the object inventory that is linked to the created list of components
2. **Bottom up**, entering object inventory first and then adding the component inventory and only then linking objects with the corresponding components
3. **Single component**, creating object inventory without creating a component list, the linking all objects to the default component
4. **Importing entire inventory** of objects and components from a pre-defined template:
   - Currently not supported. The description for this capability is forthcoming.

Top down — Create a single component and then, one by one, add object line items associated with that component. As user adds items to the inventory, considers whether to create a single object per line of inventory or group them together, and sets the unit count. When starting an estimate, a user may want to create a line item that represents multiple objects, e.g., 10 small new tables. As more information is obtained, user may want to break 10 new tables into 2 small, 5 medium and 3 complex tables. Eventually, user can even break items down further and give a descriptive name to each object in the Description column.

Bottom up — Create object inventory first by simply entering the entire list of objects. As user proceeds, the person will need to reference an existing component — select *Default-component* from the drop-down list. Then create the desired component list and link objects with the desired components by walking through the Components column in the Object Inventory table.

Single component — This is similar to the previous step — simply create objects and link them to the default component. Note that using the single component approach prevents in- out-of-scope setting from being used. This is because the default component is always in scope. In this case, the only way to take a line item out of scope is to set the unit count to 0.

Once the inventory of components and objects is set up, users can take the following actions:

- Remove/include components and corresponding line items in/out of scope
- Link/de-link components to L1 business processes
- Adjust by changing Adjustor column, e.g., when a line item is either simple or complex

- Change line-item unit count
- Change line item from *new* to *modified* or vice versa

2.5 Update complexity drivers

2.5.1 Overview

In this step, update the default complexity settings to reflect the complexity of user proposal/project. Most complexity driver settings assume a neutral position, with each complexity driver set to *medium*, causing zero percentage adjustment.

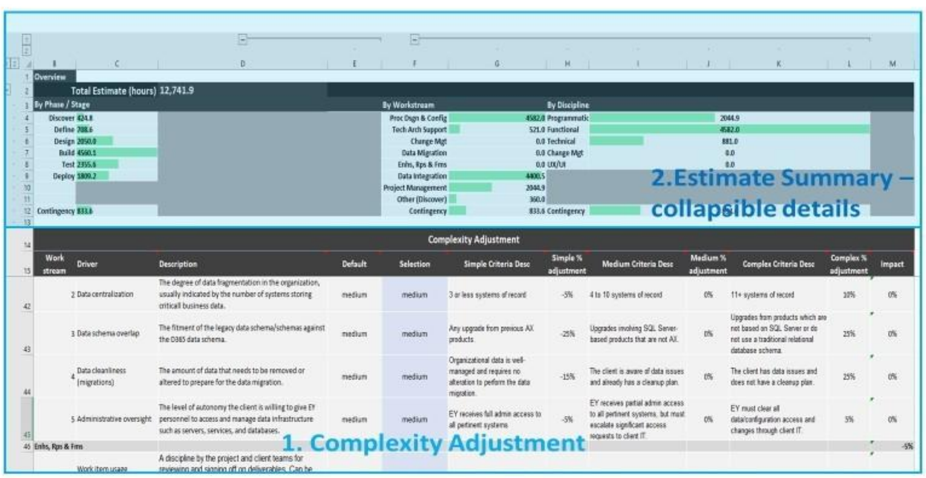
The complexity drivers are linked to their parent workstreams. The individual complexity drivers allow the increase or decrease of effort of the corresponding workstreams by a set percentage. The adjustments for individual drivers that are associated with a given workstream are summed up, and the total adjustment is applied to base effort of the corresponding workstream.

For each driver, users can select *simple*, *medium* or *complex* setting: *simple* will result in a negative adjustment percentage, *medium* in zero percentage adjustment and *complex* in a positive percentage adjustment.

Note that although users can select a level for a given complexity driver, they cannot directly adjust the percentage of adjustment.

Individual complexity driver descriptions and the selection criteria allow users to choose the appropriate complexity level. This step is carried out the Complexity worksheet.

2.5.2 Complexity worksheet details



The Complexity worksheet enables users to adjust the estimate based on the complexity of the opportunity or project being estimated. The main section is the Complexity Adjustment table alongside the Estimate Summary section on the header.

2.5.2.1 Complexity Adjustment

Work stream	Driver	Description	Default	Selection
Business Process Design & Config				
1	Process Maturity	Planned implementation process, e.g., agile: medium — started using such process on a few projects; complex — new to the process.	medium	medium
2	Multiple Countries or Regions	Differing cultures and regulatory standards can add complexity to the solution: multiple currencies, language barriers, regulatory standards.	medium	medium
3	Multiple Industries and Business Lines	Different industry regulatory requirements, multiple product lines and sales channels, nonstandard processes across the business lines, and reporting and financial consolidation complexities.	medium	medium

Simple Criteria Desc	Simple % adjustment	Medium Criteria Desc	Medium % adjustment	Complex Criteria Desc	Complex % adjustment	Impact
						0%
Has experience executing multiple projects with such process	-5%	Started using such process on a few projects	0%	New to the process	5%	0%
Single rollout and nationally based (single country or region)	-10%	Two or three phased regional rollouts	0%	More than three regional rollouts	10%	0%
Single industry	-10%	Two to five	0%	More than five	10%	0%
Has experience executing multiple projects with such process	-5%	Started using such process on a few projects	0%	New to the process	4%	0%

This table is designed for adjusting the estimation based on complexity of the project. Each complexity driver linked to a specific workstream impacts the adjustment percentage on the base effort of that workstream based on the driver level: small, medium and large:

- **Workstream** — name of each workstream that includes complexity drivers
- **Driver** — an individual complexity element that adjusts a given workstream's base effort by a certain percentage:
    - A complexity driver can be selected in one of three levels: simple, medium and complex. A medium setting does not require any adjustment to the effort (adjustment = 0%), while simple selection results in negative adjustment and complex setting results in positive adjustment to the effort. For most complexity drivers, medium is the default level.
- **Description** — detailed description of the complexity driver
- **Default** — default level setting for the driver, usually medium
- **Selection** — the desired setting for the level of a complexity driver; same as the default setting unless modified
- **Simple criteria description** — detailed explanation of why and when a simple level should be selected
- **Simple percentage adjustment** — actual adjustment to the effort if the simple level is selected for a given complexity driver; typically, a negative percentage (-%)
- **Medium criteria description** — detailed explanation of why and when a medium level should be selected
- **Medium percentage adjustment** — actual adjustment to the effort if the medium level is selected for a given complexity driver (0%)
- **Complex criteria description** — detailed explanation of why and when a complex level should be selected
- **Complex percentage adjustment** — actual adjustment to the effort if the complex level is selected

for a given complexity driver; typically, a positive percentage (+%)

- **Impact** – displays the effect by the individual complexity drivers or the total adjustment for a given workstream:
    - Calculated based on all the drivers within a workstream, and the result of that calculation is reflected on the workstream complexity percentage in the Estimate worksheet.

2.5.2.2 Estimate Summary – collapsible section

Total Estimate (hours): 615.4					
By Phase / Stage		By Workstream		By Discipline	
Discover	0	Project Management	575	Programmatic	575
Plan	108	Business Process Design & Config	0	Functional	0
Confirm & Design	0	Enhancements	0	Technical	0
Progressive Build	208	Reports	0	Change Mgt	0
Test	182	Data Integration	0	UX/UI	0
Deploy	65	Data Conversion	0		
Support	13	Change Management	0		
		Technical Arcitecture	0		
		Security & Compliance	0		
Contingency	40	Contingency	40	Contingency	40
TOTAL	615		615		615

The total effort estimate is always displayed at the top of the worksheet. However, the details normally hidden can be expanded by clicking on a "+" sign. Once expanded, this section contains estimate summary broken down by phases/stages, workstreams and disciplines. The detailed description of these is identical for all worksheets and can be found under *Estimate*.

2.5.3 Using the Complexity worksheet

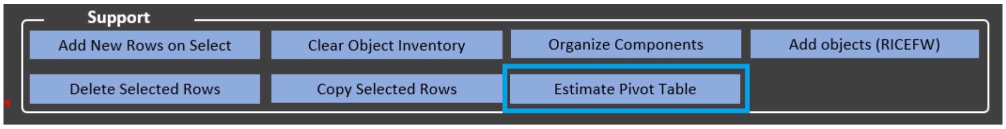
Complete the estimating step on this worksheet as follows:

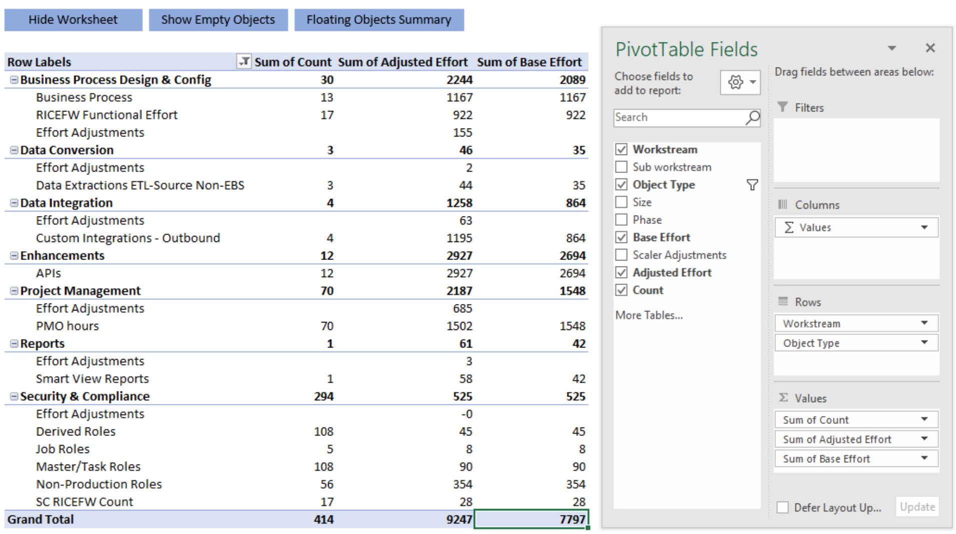
1. Before starting on the process, make a note of the total estimate. This will help the user assess the impact of the change the user will be making. Consider expanding the total section and making a screenshot – this enables the user to do before-and-after comparison to greater level of granularity.
2. Walk through this worksheet, reviewing the default setting and making the adjustments workstream by workstream.
3. For each workstream, review the individual complexity driver setting, review complexity descriptions and selection criteria, and then adjust the individual driver settings.
4. Review the total adjustment for the workstream and re-review individual driver settings to match the requirements and the context of the user proposal/project.
5. After finishing the complexity adjustment, go back and compare it to the estimate before going through the complexity adjustment process.
6. Revisit workstreams that appear to be adjusted too much in one or another direction.

2.6 Estimate Pivot Table

In the estimate pivot table user is able to review all drivers with their total amount, as well as total effort. To reveal the Estimate Pivot Table sheet user can click on relevant buttons in Functional or Components worksheet. On below figures the buttons are highlighted for respectively: Functional and Components worksheet.

Process Scope Selections			Estimate Pivot Table	Start Edit	Modules		
Scope	L1	L2	L3		Size	Adjustor	Rationale





Row Labels	Sum of Count	Sum of Adjusted Effort	Sum of Base Effort
Business Process Design & Config	**30**	**2244**	**2089**
Business Process	13	1167	1167
RICEFW Functional Effort	17	922	922
Effort Adjustments		155	
Data Conversion	**3**	**46**	**35**
Effort Adjustments		2	
Data Extractions ETL-Source Non-EBS	3	44	35
Data Integration	**4**	**1258**	**864**
Effort Adjustments		63	
Custom Integrations - Outbound	4	1195	864
Enhancements	**12**	**2927**	**2694**
APIs	12	2927	2694
Project Management	**70**	**2187**	**1548**
Effort Adjustments		685	
PMO hours	70	1502	1548
Reports	**1**	**61**	**42**
Effort Adjustments		3	
Smart View Reports	1	58	42
Security & Compliance	**294**	**525**	**525**
Effort Adjustments		-0	
Derived Roles	108	45	45
Job Roles	5	8	8
Master/Task Roles	108	90	90
Non-Production Roles	56	354	354
SC RICEFW Count	17	28	28
Grand Total	**414**	**9247**	**7797**

This is a helpful tool for many types of summaries that user can expect. After selecting a cell within a pivot table, a control panel on the right side will appear, where user can select fields to be visible in the table in columns and rows as labels, and values as a core of the table. In the example above, the objects are broken down into each Workstream, and for each object there is its total count (every size), base effort and adjusted effort.

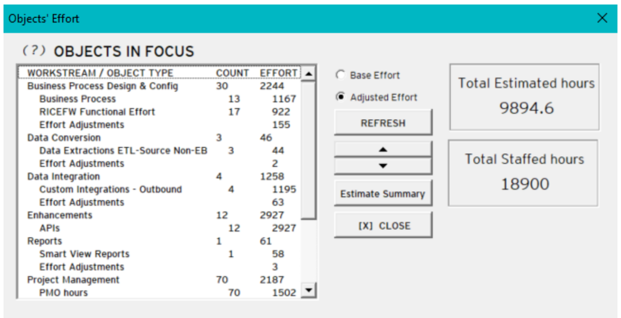
Effort Adjustments objects – this is a pseudo object that contains all additional effort that comes from specific, custom computations, related to specific stage of the project.

Hide Worksheet – when work with the pivot table is finished, user can hide worksheet to get more lean version of the estimator. It can be re-invoked after clicking triggering button as in the first place.

Show/Hide Empty Objects – user can choose if objects that have no effort should be visible (to see every available driver in the estimator) or not.

Floating Objects Summary – floating window showing all active objects. Please refer to the next subchapter

2.6.1 Floating Objects Summary



Floating Objects Summary is a floating window that contains a list of active objects sorted by their Workstreams. Active objects means objects that have non zero effort. In the list user can see the total count of current objects as well as their effort. There is a possibility to switch between two types of effort: Base and Adjusted. This can be achieved by selecting respective option box next to the list.

Floating window is always on top of the screen, so user can easily switch between different worksheets of the Estimator.

Refresh – clicking this button updates the list after some changes made by the user within the estimate

Arrows – extends or shortens the window to adjust to the needed length of the list. This is helpful when having large amount of objects, there is a need to see all of them at once.

Estimate Summary – this button redirects to another floating estimate summary window.

Close – closes the floating window

2.7 Review estimate summary

2.7.1 Overview

In this step, review the estimate from different perspectives, looking for inconsistencies and out-of-scope deviations either to a positive or negative direction:

- Workstream view lets users see the estimate broken down by workstreams
- Phase/stage view lets user see the estimate broken down by stages of work
- Discipline view lets user see the estimate broken down by Digital Delivery Methodology disciplines

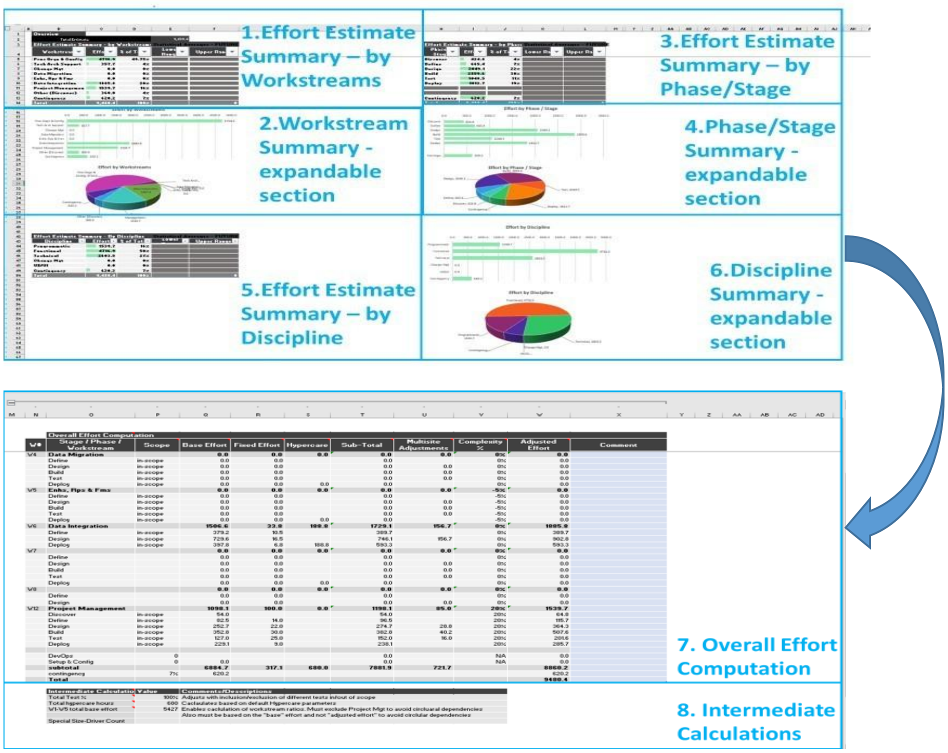
Graphs are provided to enhance visualization of effort distribution. Both bar graphs and pie charts are available.

Also, a hidden (collapsed) section contains a tabular representation of the estimate that may be helpful in some circumstances.

This is step is primarily focused on review and analysis activities, while adjustments can be made by going back to the previous steps.

This step is carried out by completing the Estimate worksheet.

2.7.2 Estimate worksheet details



This worksheet provides a breakdown of estimate by workstreams, stages/phases and disciplines. It also contains a set of calculations that represents the estimating engine:

- **Effort Estimate Summary – by Workstreams** – breaks down total estimated effort by individual workstreams and their total sum
- **Workstream Summary – expandable section** – summary breakdown via pie chart and graph
- **Effort Estimate Summary – by Phase/Stage** – breaks down total estimated effort by individual phases/stages and their total sum
- **Phase/Stage Summary – expandable section** – summary breakdown via pie chart and graph
- **Effort Estimate Summary – by Discipline** – breaks down total estimated effort by individual disciplines and their total sum
- **Discipline Summary – expandable section** – summary breakdown via pie chart and graph

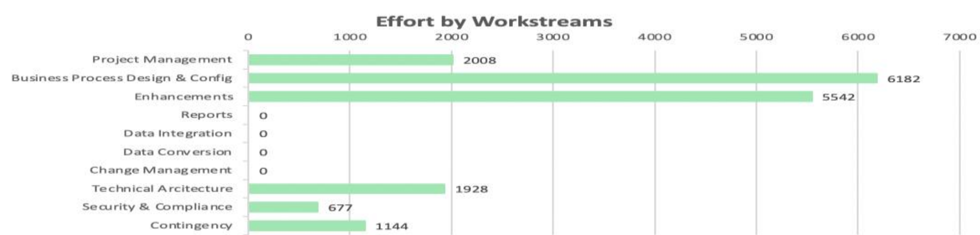
2.7.2.1 Effort Estimate Summary — by Workstreams

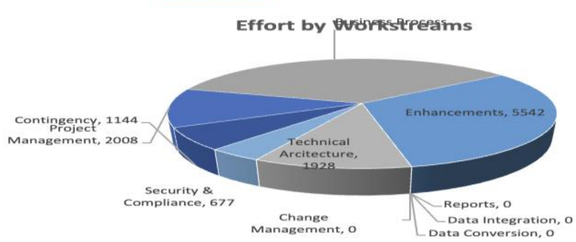
Effort Estimate Summary - by Workstreams			Statistical Averages - FUTURE	
Workstream	Effort	% of Total	Lower Range	Upper Range
Project Management	2008	11%		
Business Process Design & Confi	6182	35%		
Enhancements	5542	32%		
Reports	0	0%		
Data Integration	0	0%		
Data Conversion	0	0%		
Change Management	0	0%		
Technical Arcitecture	1928	11%		
Security & Compliance	677	4%		
Contingency	1144	7%		
Total	17481	100%		0

- **Workstream —** name of every workstream listed
- **Effort —** estimated effort associated with individual workstreams in hours
- **Percentage of total** — represents the ratio of the workstream's effort relative to the total effort, including contingency
- **Lower range** – (future capability) the lower limit of percentage of total for a given workstream determined statistically based on actual data
- **Upper range** – (future capability) the upper limit of percentage of total for a given workstream determined statistically based on actual data

2.7.2.2 Workstream Summary — expandable section

This section consists of a graph and a pie chart that visually summarize effort estimates by workstreams.





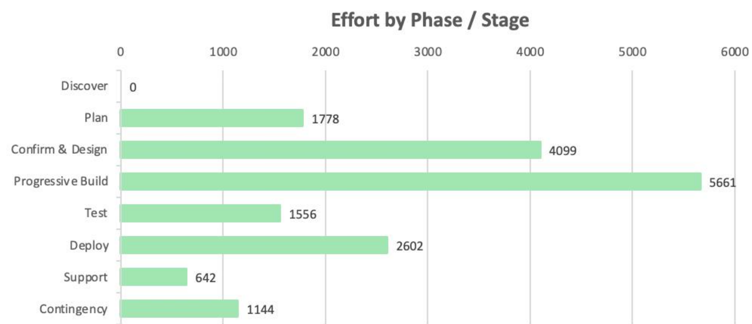
2.7.2.3 Effort Estimate Summary — by Phase/Stage

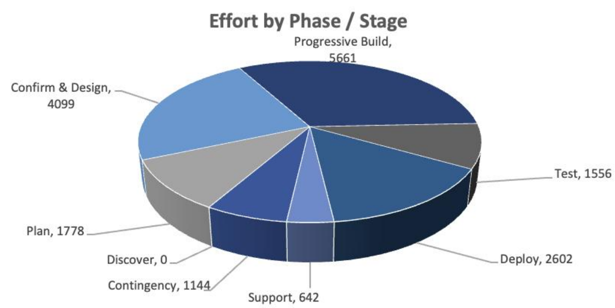
Effort Estimate Summary - by Phase/Stage			Statistical Averages - FUTURE	
Phase / Stage	Effort	% of Total	Lower Range	Upper Range
Discover	0	0%		
Plan	1778	10%		
Confirm & Design	4099	23%		
Progressive Build	5661	32%		
Test	1556	9%		
Deploy	2602	15%		
Support	642	4%		
Contingency	1144	7%		
Total	17481	100%		0

- **Phase/stage** — name of every phase/stage listed
- **Effort** — estimated effort associated with individual phase/stages in hours
- **Percentage of total** — represents the ratio of the phase's/stage's effort relative to the total effort, including contingency
- **Lower range** — (future capability) the lower limit of percentage of total for a given phase/stage determined statistically based on actual data
- **Upper range** — (future capability) the upper limit of percentage of total for a given phase/stage determined statistically based on actual data

2.7.2.4 Phase/Stage Summary — expandable section

This section consists of a graph and a pie chart that visually display effort estimate summary by phase/stage.





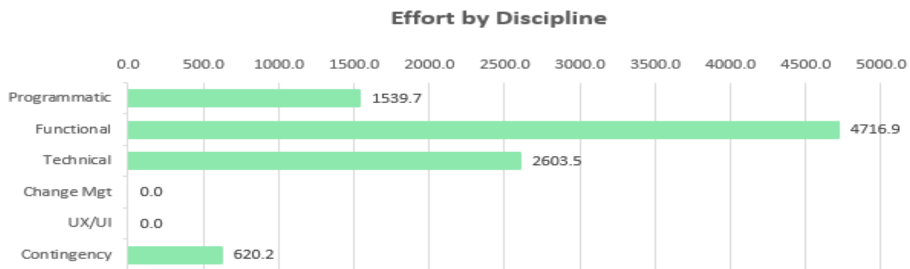
2.7.2.5 Effort Estimate Summary — by Discipline

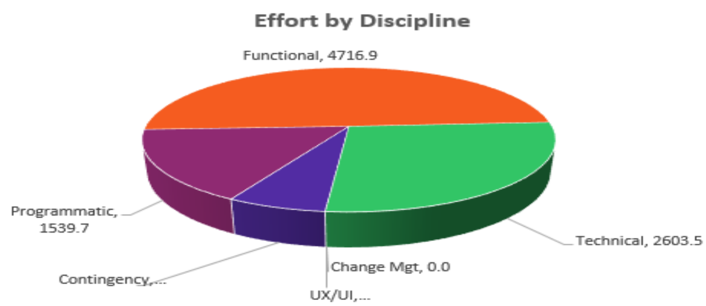
Effort Estimate Summary - By Discipline			Statistical Averages - FUTURE	
Discipline	**Effort**	**% of Total**	**Lower Range**	**Upper Range**
Programmatic	**1539.7**	**16%**		
Functional	**4716.9**	**50%**		
Technical	**2603.5**	**27%**		
Change Mgt	**0.0**	**0%**		
UX/UI	**0.0**	**0%**		
Contingency	**620.2**	**7%**		

- **Discipline —** name of every discipline listed
- **Effort —** estimated effort associated with individual discipline in hours
- **Percentage of total** – ratio of the discipline's effort relative to the total effort, including contingency
- **Lower range** – (future capability) the lower limit of percentage of total for a given discipline determined statistically based on actual data
- **Upper range** – (future capability) the upper limit of percentage of total for a given discipline determined statistically based on actual data

2.7.2.6 Discipline Summary — expandable section

This section consists of a pie chart and bar graph that visually display effort estimate summary by discipline.





2.7.2.7 Overall effort computation

This table represents the estimating engine of this workbook. The general calculation and computation occur in this table, including all workstreams, phases and disciplines.

20	**Project Management**		**2008.1**	**0.0**	**2008.1**	**0.0**	**0%**	**2008.1**	
	Discover	out-of-scope	0.0	0.0	0.0			0.0	
	Plan	in-scope	259.5	0.0	259.5		0%	259.5	
	Confirm & Design	in-scope	372.6	0.0	372.6	0.0	0%	372.6	
	Progressive Build	in-scope	703.7	0.0	703.7	0.0	0%	703.7	
	Test	in-scope	306.9	0.0	306.9	0.0	0%	306.9	
	Deploy	in-scope	295.2	0.0	295.2		0%	295.2	
	Support	in-scope	70.2	0.0	70.2		0%	70.2	
	subtotal		**16099.6**	**237.9**	**16337.5**	**0.0**		**16337.5**	
	contingency	7%						1143.6	
	Total							**17481.1**	

- **Workstream number** — ID for listed workstream/phase/stage
- **Stage/phase/workstream** — list of workstreams and elements within those workstreams
- **Scope** — status per workstream based on the scope selection process in Scope worksheet
- **Base effort** — an effort calculated by summing up objects that are associated with a given workstream:
    - In some workstreams, such as project management, the base effort is calculated as a percentage of other workstreams.
- **Fixed effort** — work associated with individual stages within individual workstreams, then added

to the base effort to form the subtotal for a given stage within a given workstream

- **Subtotal** – sum of base effort, fixed effort and hypercare per stage
- **Geographic distribution adjustments** – additional overhead effort associated with individual phases of work within the specific workstream
- **Complexity percentage** – adjustment associated with a given workstream (see Complexity worksheet)
- **Adjusted effort** – based on subtotal, with the addition of geographic distribution adjustment, and adjusted by complexity percentage:
    - (Calculation – [(Sub-Total + Geographic distribution Adjustments) *1+ (Complexity %)])
- **Comment** – any notes or comments

2.7.2.8 Intermediate Calculations

Intermediate Calculations	Value	Comments/Descriptions
Total Test %	100%	Adjusts with inclusion/exclusion of different tests in/out of scope
W1-W19 total base effort	14092	Enables caclulation of workstream ratios. Must exclude Project Mgt to avoid circluaral dependencies. Also must be based on the "base" effort and not "adjusted effort" to avoid circlular dependencies
Total RICEFW	0	Total number of all tech objects small, medium, large (new, modified)
Total L3 Business Processes	643	Total L3 Business process small, medium, large
Base Discover Hours	0	if at least one workstream has discover phase in scope then base discover calculates based off parameters
Base Sustain Hours	0	if at least one workstream has sustain phase in scope then base discover calculates based off hyoercare parameters
Total Hours Without Sustain/Discover	13757.9	Total hours of all workstreams excluding sustain and discover

- **Intermediate calculations** – total test percentage, total hypercare hours, etc.
- **Value** – given percentage for individual calculation
- **Comments/descriptions** – explanation of the calculations and any additional comments

2.7.3 Using the Estimate worksheet

The review and analysis of the estimate can be carried out in a couple of different ways. A thorough analysis may require using more than one review method below.

Review and analysis by workstream:

1. Remove all stages and workstreams from scope.
2. Continue analysis on a workstream-by-workstream basis.
3. Bring the first workstream that is not project management into scope.
4. Review individual workstream effort by stage of work.
5. Disable the workstream user have reviewed.
6. Move on to review the next workstream.

Keep in mind the dependencies between workstreams. For example, the Technical Architecture Support workstream will depend on all other technical workstreams. Thus, users cannot review the Technical Architecture Support workstream individually – they need to include the dependent workstreams in scope. Similarly, the Project Management workstream depends on all other workstreams, and users cannot review it individually either. Users can only review it after all other workstreams are included in scope.

As user walks through the individual workstreams, look for outsize/undersize effort in each stage within a workstream. Should any discrepancies be found, users should go back to the previous steps and look at the size driver and the complexity driver setting for workstream in focus.

Review and analysis by stage

The process is very similar here, except the role of workstreams and stages is reversed:

1. Remove all stages and workstreams from scope.
2. Continue analysis on a stage-by-stage basis.
3. Bring the first stage into scope.
4. Review individual stage effort on workstream-by-workstream basis.
5. Disable the stage user has reviewed.
6. Move on to review the next stage.

Lastly, user may review the effort by disciplines — review the Summary by Discipline table.

For additional insights, see the Overall Effort Computation table that is collapsed on the right-hand side of the worksheet.

3 Steps for creating resource plan

As users move from the effort estimate into a resource plan, they should compare the financial model for the effort to the staffing plan. General alignment and comparing the estimate and the resourcing plan are critical to producing an overall quality estimate. During the process, users should focus on estimated and staffed hours financials located at the top of the Resources worksheet along with Cost and Price and Expenses worksheets for the overall computation.

The resource planning engine comprises four major sections:

1. Create initial resource plan
2. Update financial parameters
3. Refine and make adjustments
4. Review indicative financials

3.1 Revisiting assumptions

Revisiting assumptions throughout the process is ideal to confirm that they are represented within the Estimator. Verifying duration and starting date along with complexity drivers helps produce a higher quality estimate. Validating correct rates are used increases accuracy of key financial metrics.

3.2 Create initial resource plan

3.2.1 Overview

The purpose of this step is to create an initial resource plan that closely aligns with the effort estimate created earlier in the estimation process. The process of creating the resource plan can be semiautomated or manual.

The semiautomated method allows the user to quickly build up the resource plan by selecting one or more pre-configured teams with all parameters that are required by the resource plan. Multiple teams are available, including either DDM-based teams or the teams specific to a given practice domain, such DET, MSFT and DDnA. Further, in many cases, teams of different sizes are available to accommodate different scenarios. Each team comprises a set of roles with assigned standard resource parameters such as level, workforce, cost, etc.

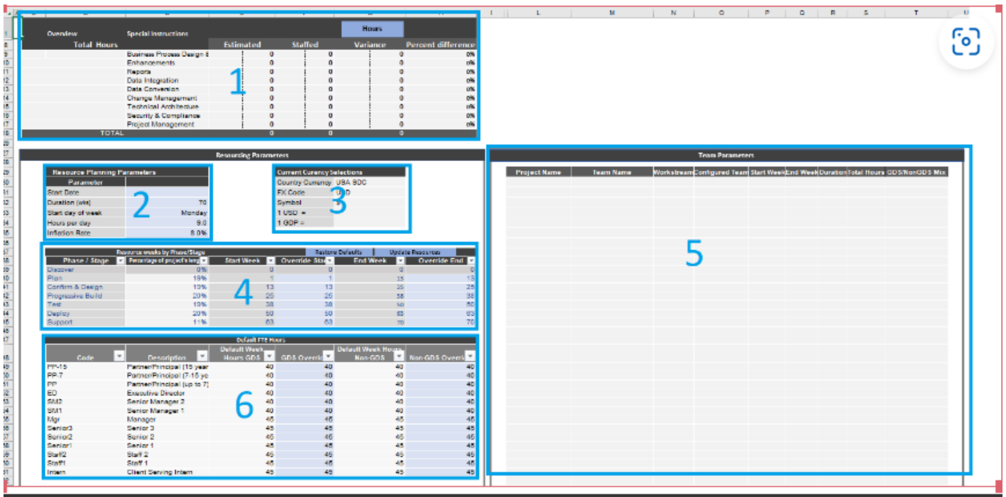
As the resource plan is constructed, the user can observe the indicative financial metrics that are continuously refreshed and provide accurate indication as to cost, prices/revenue, margin and EAF. The notion of "indicative" is used to indicate that the formal process of computing financials still requires working through the Mercury application.

Once the first-cut resource plan is established, it can be adjusted and fine-tuned through a manual process by updating FTEs/hours provided with the plan, by adding/deleting roles or by adjusting role parameters. Fine-tuning the resource plan is aided by a dashboard that allows quick comparison between estimated and staffed hours at the total or as well at a more granular level.

Of course, a fully manual process for creating a resource is also supported. In a manual process, a team name ought to be created and all added resource will be a part of that team. Each resource must be added with all required parameters such as level, workforce, fiscal year, etc. In all other aspects, the manual entry is similar to the semiautomated process: (1) indicative financials are continuously computed and (2) an alignment procedure is used to facilitate a close match between estimated and staffed hours.

All functions described above are implemented on the Resources worksheet.

3.3 Resources Parameters Worksheet in Detail



Below is an in-depth analysis of the features in our Resources Parameters Tab:

1. **Overview:**
Providing an overarching view of the hours outlined in the project, this feature incorporates the computed total of Estimated, Staffed, and Variance hours. It's an effective way to get a comprehensive grasp of the project's current time status.

2. **Resource Planning Parameters:**

- **Start Date:** The beginning of your project.
- **Duration:** The project's lifespan in weeks.
- **Start Day of the week:** The week's start day.
- **Hours per Day:** The daily working limit.
- **Inflation Rate:** The expected inflation rate.

3. **Current Currency Selections:**

- **Country Currency:** The currency of the selected country.
- **FX Code:** The foreign exchange code.
- **Symbol:** The currency symbol.
- **1 USD and 1 GDP:** Conversion rates for reference.

4.**Resource Weeks by Phase/Stage:**

This table breaks down the project length across different phases/stages, with the relevant weeks and an option to override start and end weeks.

1. **Team Parameter:** lists all teams added in the Resources worksheet

E.g.

3.3.1.1 Team parameters

Team Parameters							
Team Name	Workstream	Configured Team	Start Week	End Week	Duration	Total Hours	GDS/NonGDS Mix
MSERP		ERPLarge	1	260	10	137722	Standard

- **Team name** – name of the teams added in the Resources worksheet
- **Workstream** – field of activity for each team
- **Configured team** – the name of pre-configured team that this team includes
- **Start week** – beginning week of the engagement
- **End week** – last week of the engagement
- GDS/Non GDS mix refer this section 3.3

Commented [AM1]: Ptm change

6. **Default FTE Hours:**

The default week hours and their overrides are listed for different codes indicating diverse job roles, both for GDS and Non-GDS.

Commented [AM2]: @Jakub Kątski@David Falcone@Matt J Nekic@Mark Gurevich

Default FTE Hours					
Code	Description	Default Week Hours GDS	GDS Override	Default Week Hours Non-GDS	Non-GDS Override
PP-15	Partner/Principal (15 years+)	40	40	40	40
PP-7	Partner/Principal (7-15 years)	40	40	40	40
PP	Partner/Principal (up to 7)	40	40	40	40
ED	Executive Director	40	40	40	40
SM2	Senior Manager 2	40	40	40	40
SM1	Senior Manager 1	40	40	40	40
Mgr	Manager	45	45	45	45
Senior3	Senior 3	45	45	45	45
Senior2	Senior 2	45	45	45	45
Senior1	Senior 1	45	45	45	45
Staff2	Staff 2	45	45	45	45
Staff1	Staff 1	45	45	45	45
Intern	Client Serving Intern	45	45	45	45

Default FTE Hours Feature /Default FTE hours Split by GDS/Non-GDS (All Levels)

Our model includes an innovative Default FTE Hours feature to optimize workforce management. Let's explore how to use this feature effectively:

Default FTE Hours Feature

This feature includes a comprehensive table that provides the default week hours for each level in your organization with specific GDS and non-GDS roles. It also enables you to input an override value if there's a deviation from the norm.

Each row in the table corresponds to a different job role or level, identified by a unique code and description. For each level coded and described, you find two pairs of columns:

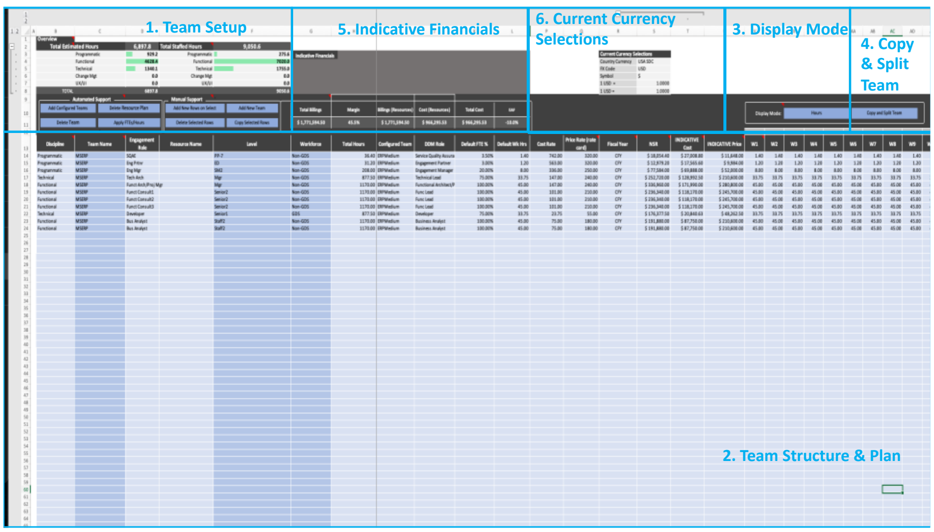
- One pair (Default Week Hours GDS and GDS Override) corresponds to GDS roles.
- The other pair (Default Week Hours Non-GDS and Non-GDS Override) corresponds to non-GDS roles.

For each role, you can view the default week hours in the relevant column, and if there's been an adjustment, you can see the overridden value in the Override column.

For instance, the 'Manager' role coded as 'Mgr' has default week hours set at 45 for both GDS and non-GDS designations. If the week hours for this role have been overridden due to special circumstances, the new adjusted hours would be shown in the respective Override column.

The Default FTE Hours feature provides a panoramic view of your workforce's hour allocation at each hierarchical level. This enhances your ability to manage your employee resources more effectively, ensuring the best use of the workforce based on their roles and ensuring organizational productivity.

3.3.2 Resources worksheet details



One of the crucial components of the resource planning is the team structure for Microsoft and DDM practices. While Microsoft practice is broken into three ERP teams (small, medium and large), Digital Deliver Methods has two cross-discipline teams (small and large) and three scrum teams (small, large and GDS). On the Team Config button, the user will have access to these teams' structure, including resource's role, level, resource type, discipline and subdiscipline, GDS (Global Delivery Services) status, and default FTE percentage.

For more accurate planning, user will be able to add individual teams in Resources worksheet by specifying team name, start week and end week. Once a team is added, resource's default FTE percentage, default working hours in a week, cost rate and total hours are automatically calculated based on their role, level and GDS status. There is another functionality where users will be able to copy and split team's timeline so that the total hours do not change. It provides support for multiple releases with slight variations in roles and resource level and longer-running engagement to allow changes in resource costs due to promotions.

When the display mode is on FTE, users can click on "Apply FTEs/Hours" to project FTE percentage for each resource based on the data located Default FTE % column in the Team Config worksheet. Users can also switch the display mode to FTE if it is in hours mode and accumulate all the hours for total staffed hours calculation. By doing so, this planning worksheet provides a distinct narrative between total estimated hours versus total staffed hours and their total cost and billings.

Commented [AM3]: Default FTE hours Split by GDS/Non-GDS (All Levels) @Jakub Kątski@David Falcone@Mark Gurevich

Commented [DF4R3]: @Amit Mishra Can you please edit this to use the formatting from other sections? E.g. use bold format on "Default FTE Hours Feature /Default FTE hours Split by GDS/Non-GDS (All Levels)" and triangle bullets for items to be defined and circle bullets for sub sections (see section 2.5.2 Component details for an example of this). Also reduce the amount of white space. Use these suggestions in general throughout userguide editing so our sections are consistent.

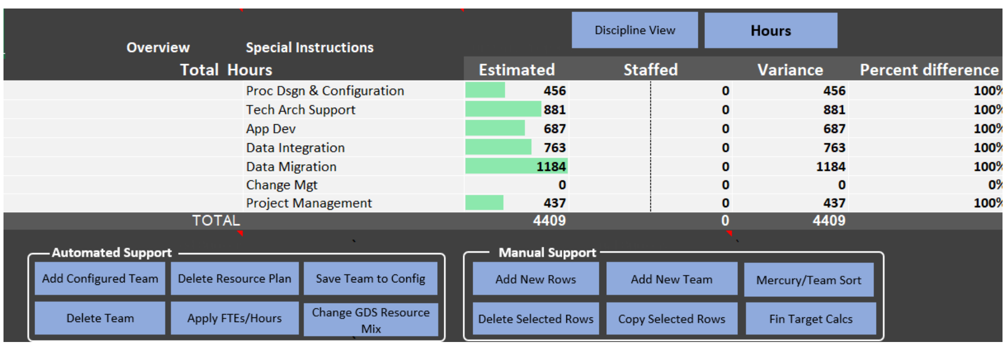
Commented [DF5R3]: Accepted

These calculations are predominantly based on the cost rate, NSR and rate card values mentioned in the Cost and Price worksheet. These prices vary between current fiscal year (CFY) and then by the next three fiscal years (CFY+1, CFY+2 and CFY+3). While calculating for indicative cost and indicative price that are based on cost rate and pricing rate, respectively, users can manually verify the cost rate and rate card values for each resource based on their level and GDS status for a specific fiscal year. By doing so, they can modify the resource plan to gain balance between estimated hours and staffed hours and total cost and billings for each.

This worksheet covers 6 major sections:

1. **Team setup** – portrays how to add pre-configured teams, add new teams from scratch, delete a team, delete the entire plan and apply FTEs/hours in the resource plan
2. **Team structure and plan** – shows the entire organization of a team being added in the plan
3. **Display mode** – users can display this table based on hours or FTEs
4. **Copy and split team** – allows users to copy and split a team that already exists in the table
5. **Indicative financials** – calculation of the resource plan
6. **Current Currency Selections** - present country's currency selections

3.3.2.1 Team setup



- **Total estimated hours** – total effort estimate in hours taken from the Estimate worksheet
- **Total staffed hours** – total hours calculated by summing hours for individual resources throughout the project
- **Add configured teams** – allows user to add a pre-configured team to the worksheet
- **Delete team** – allows user to delete a specific team from the worksheet
- **Delete resources plan** – allows user to the delete the entire plan (all resources) from the worksheet
- **Apply FTEs/hours** – plots FTEs across the specified months for the specified team already included in the resource plan
- **Save Team to Config** - saves chosen team from canvas as a Configured Team so that user can add it in the future clicking on "Add Configured Team"
- **Add new team** – allows user to add a new team that is not pre-configured
- **Add new rows on select** – lets user add single or multiple rows from a selected point (e.g., cell B5)
- **Delete selected rows** – deletes selected rows from the resource plan

- **Copy Selected Rows** – works exactly as the button in Components worksheet where user can copy and paste selected rows into the resource tables
- **Mercury/Team Sort** – it allows user to sort the table primarily based on the Workforce status in the following order: Non-GDS, GDS, Client and Contractors. On the other hand, default view will sort the table based on teams that are created in order
- **Fin Target Calculations** – allows user to modify following financial targets to accommodate change requests more efficiently: 1) Fee (billings), 2) Margin % and 3) Blended Rate. User can provide expected values for these items and overall resource estimation will alter accordingly including bill rates for individual resources.

Change GDS Resource Mix -

  - GDS/Non-GDS mixes across teams in resource canvas
  - User Guide: Workforce Type and Resource Mix Configuration
  - Our system optimizes diverse team configurations by assigning each team with a default workforce type of either Global Delivery Services (GDS) or non-GDS.
- **Workforce Types:**
  - Always GDS: Unless the selection is 'All Non-GDS', the assigned resource remains GDS.
  - GDS or NonGDS: The assigned resource oscillates between GDS and Non-GDS based on the level of aggressiveness chosen for the project.
  - Always NonGDS: Regardless of the selection, these resources remain Non-GDS.
  - Manipulating and managing team configurations hinge upon two key concepts: Resource Allocation and Resource Mix Levels.
- **Resource Allocation:**
  - Adding a Team: When adding a team, the system automatically allocates GDS or Non-GDS resources based on the chosen aggressiveness level.
  - Changing Settings: If you adjust the aggressiveness level, every team resource automatically updates based on their assigned workforce type.
  - Manual Override: Users can manually change the assigned workforce type. However, note that any adjustment to the aggressiveness level will override these changes.
- **Resource Mix Levels:**
- The system provides five configurable resource mix levels:
- All Onshore: Regardless of their originally assigned type, all resources will be set to Non-GDS.
  - Safe: All resources will transition to Non-GDS except for the ones under 'Always GDS'.
  - Standard: Resources conform to their default workforce type as per the pre-configured team assignment.
  - Aggressive: All resources will switch to GDS except for those labeled as 'Always Non-GDS'.
  - All GDS: Every assigned resource switches to GDS, with no exceptions.

Commented [AM6]:

Commented [AM7R6]:

Commented [AM8R6]: @Jakub Kątski@David Falcone@Mark Gurevich

Commented [JK9R6]: @Amit Mishra why this is here and not under 3.2.1.1?

Commented [AM10R6]: Sure @Jakub Kątski moving to 3.2.1.1

Commented [DF11R6]: @Amit Mishra Can you please change the formatting to be consistent with the rest of the document? There shouldn't be triangle bullets on every line, descriptions under the **bold** sections should be indented (eg the numbers under **Worksforce Types:**), remove unnecessary white space (review other sections for examples), and remove "-" bullets and use either numbers or the triangle/ circle bullets found in other sections to stay consistent.

Commented [DF12R6]: Accepted

3.3.2.2 Team structure and plan

Discipline	Team Name	Engagement Role	Resource Name	Level	Workforce	Workstream	Total Hours
Programmatic	MSERP	Eng Prtnr		ED	Non-GDS		478.00
Programmatic	MSERP	Eng Mgr		SM2	Non-GDS		168.00
Technical	MSERP	Tech Arch		Mgr	Non-GDS		450.00
Functional	MSERP	Funct Arch/Proj Mgr		Mgr	Non-GDS		190.00
Functional	MSERP	Funct Consult1		Senior2	Non-GDS		1170.00
Functional	MSERP	Funct Consult2		Senior2	Non-GDS		1170.00
Technical	MSERP	Developer		Senior1	GDS		585.00
Functional	MSERP	Bus Analyst		Staff2	Non-GDS		1170.00

Configured Team	DDM Role	Default FTE %	Default Wk Hrs	Override Hours	Total Hours	INDICATIVE Cost	INDICATIVE Price	Margin %	Ramp Up (in Weeks)	Ramp Down (in Weeks)	Ramp up Hours	Ramp down Hours

Cost Rate	Price Rate (rate card)	Fiscal Year	NSR	INDICATIVE Cost	INDICATIVE Price	W1	W2	W3	W4
608.00	320.00	CFY	$198,848.00	$290,624.00	$152,960.00	50.00	50.00	50.00	50.00
363.00	250.00	CFY	$62,664.00	$60,984.00	$42,000.00	8.00	8.00	8.00	8.00
159.00	240.00	CFY	$129,600.00	$71,550.00	$108,000.00	0.00	0.00	0.00	0.00
159.00	240.00	CFY	$54,720.00	$30,210.00	$45,600.00	5.00	5.00	5.00	5.00
109.00	210.00	CFY	$236,340.00	$127,530.00	$245,700.00	45.00	45.00	45.00	45.00
109.00	210.00	CFY	$236,340.00	$127,530.00	$245,700.00	45.00	45.00	45.00	45.00
26.00	55.00	CFY	$117,585.00	$15,210.00	$32,175.00	22.50	22.50	22.50	22.50
81.00	180.00	CFY	$191,880.00	$94,770.00	$210,600.00	45.00	45.00	45.00	45.00

- **Discipline** – name of each discipline individual resource falls into
- **Team name** – an assigned team name which by default, derives from the corresponding pre-configured team name
- **Engagement role** – the name, derived from the DDM role, of the function assigned within a given team and engagement context; must be unique
- **Resource name** – The first and last name of a specific resources assigned to the role:
  - By default, the level of the assigned person will match the expected level for the role.
- **Level** – The EY rank associated with a given role
- **Workforce** – indicates the resource source for a given role; can be non-GDS, GDS, contractor or client
- **Workstream** - link that resource and their hours to a specific workstream
- **Total hours** – the total staff hours for a given role/resources for the entire engagement
- **Configured team** – name of the preconfigured team automatically chosen upon adding a team
- **DDM role** – standard name picked from the list of role names defined within DDM; does not have to be unique on a team
- **Default FTE %** – a value from 0% to 100% that represents the portion of time a given resource will be assigned to the role
- **Default week hours** – based on level/location, the default weekly working hours for a given role
- **Override Hours** - if this cell is not empty it will override Default week hours
- **Ramp Up (in Weeks)** - allows user to enter a numerical number to be given specific hours for an individual resources from the start of the project duration (i.e., if ramp up in weeks is 5, that means it will impact the first 5 weeks of that resource)

- **Ramp Down (in Weeks)** - allows user to enter a numerical number to be given specific hours for an individual resources from the end of the project duration (i.e., if ramp down in weeks is 5, that means it will impact the last 5 weeks of that resource)
- **Ramp up Hours** – these hours are only applicable when Ramp up in weeks is provided with a value (i.e., if ramp up in weeks is 5 and ramp up hours is 10, that means for that individual resource, the first 5 weeks will have 10 hours each regardless of the default FTE% and default week hours)
- **Ramp down Hours** – these hours are only applicable when Ramp down in weeks is provided with a value (i.e., if ramp down in weeks is 5 and ramp down hours is 10, that means for that individual resource, the last 5 weeks will have 10 hours each regardless of the default FTE% and default week hours)
    - One key distinction between ramp up/down weeks/hours is that when weeks overlap, ramp down weeks and hours override ramp up weeks and hours
- **Cost rate** — price for a given role/resource — applies **only** to this resource; depends on the fiscal year selection for the specific role
- **Price rate (rate card)** — same as rate card value from Cost and Price worksheet based on workforce type and fiscal year
- **Fiscal year** — during which the specific role/resource is considered
- **NSR** — net service revenue associated with a specific role based on workforce type and fiscal year; calculated with the NSR price from Cost and Price worksheet, then multiplied by the total hours for that resource in the engagement
- **Indicative cost** — total cost of the specific role/resource across the entire engagement
- **Indicative price** — total price of the specific role/resource across the entire engagement
- **Week 1—week N** — contains either hours or FTE percentage for a given role/resource in each week, where N is the week number starting from the beginning of the engagement

3.3.2.3 Display mode

Hours

Display Mode: Hours

	4-Feb	11-Feb	18-Feb	25-Feb	4-Mar	1
ce	W1	W2	W3	W4	W5	
.08	45.0	45.0	45.0	45.0	45.0	
.08	45.0	45.0	45.0	45.0	45.0	
.56	45.0	45.0	45.0	45.0	45.0	
.76	22.5	22.5	22.5	22.5	22.5	
.34	22.5	22.5	22.5	22.5	22.5	
.52	45.0	45.0	45.0	45.0	45.0	
.04	8.0	8.0	8.0	8.0	8.0	
.50	1.4	1.4	1.4	1.4	1.4	

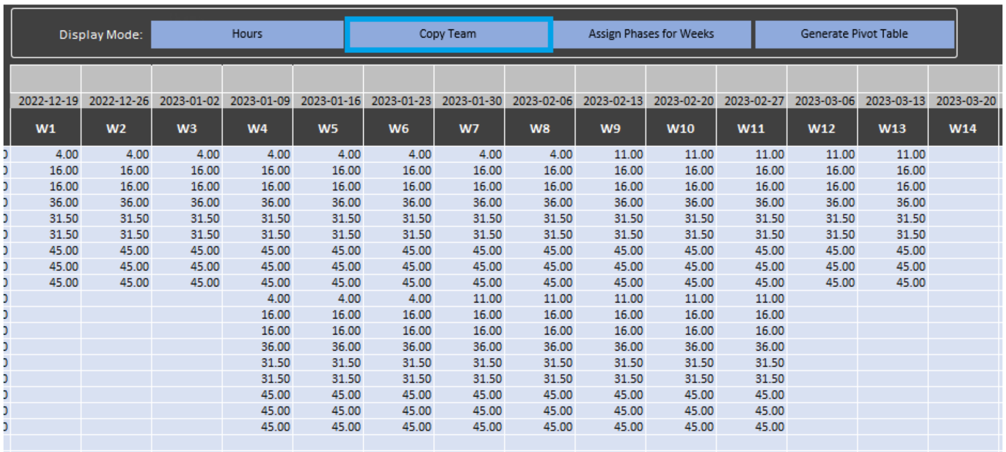
FTE

Display Mode: FTE

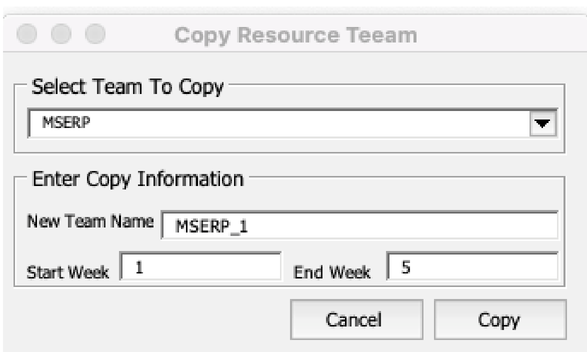
4-Feb	11-Feb	18-Feb	25-Feb	4-Mar	1
W1	W2	W3	W4	W5	
1.0	1.0	1.0	1.0	1.0	
1.0	1.0	1.0	1.0	1.0	
1.0	1.0	1.0	1.0	1.0	
0.5	0.5	0.5	0.5	0.5	
0.5	0.5	0.5	0.5	0.5	
1.0	1.0	1.0	1.0	1.0	
0.2	0.2	0.2	0.2	0.2	
0.0	0.0	0.0	0.0	0.0	

- **Hours** — displays the table in hours per week
- **FTE** — displays the table in FTEs, e.g., 0.5 or 1 corresponding with the default FTE percentage

3.3.2.4 Copy team

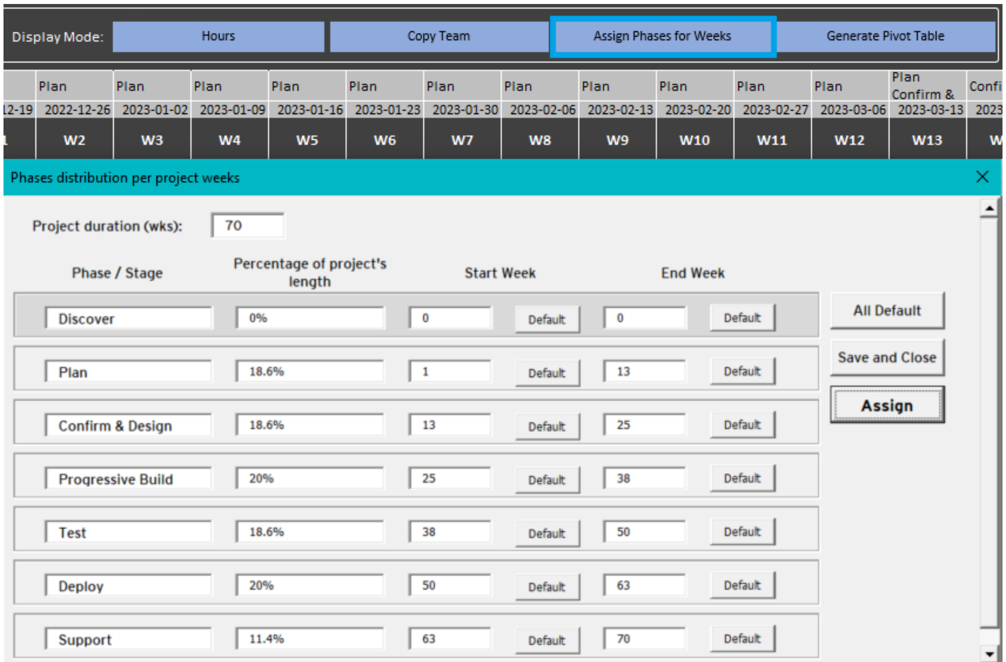


2022-12-19	2022-12-26	2023-01-02	2023-01-09	2023-01-16	2023-01-23	2023-01-30	2023-02-06	2023-02-13	2023-02-20	2023-02-27	2023-03-06	2023-03-13	2023-03-20
W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	11.00	11.00	11.00	11.00	11.00	
16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	
16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00	
36.00	36.00	36.00	36.00	36.00	36.00	36.00	36.00	36.00	36.00	36.00	36.00	36.00	
31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	
31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50	
45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	
45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	
45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00	
			4.00	4.00	4.00	11.00	11.00	11.00	11.00	11.00			
			16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00			
			16.00	16.00	16.00	16.00	16.00	16.00	16.00	16.00			
			36.00	36.00	36.00	36.00	36.00	36.00	36.00	36.00			
			31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50			
			31.50	31.50	31.50	31.50	31.50	31.50	31.50	31.50			
			45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00			
			45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00			
			45.00	45.00	45.00	45.00	45.00	45.00	45.00	45.00			



This button allows user to copy a selected team depending on precise week from where the table should be applied. For example, if a team is added with the duration from week 1–13, then, via this button, this team can be copied multiple times into selected range of duration maintaining the Ramp Up/Down original settings. Furthermore, those new teams created due to the copy functionality can also be copied into further sections for a better planning estimate.

3.3.2.5 Assign Phases for Weeks



This Assign Phases for Weeks button allows user to schedule the Phases/Stages of the project within each week while preparing resources for it. User is able to modify Start Week and End Week of each Phase. There is a data validation within the entry fields that enables user to submit only positive numbers, where Start Week is lower or equal to End Week of current Phase. After changing values, user can always revert to the default one by clicking a 'Default' button on the right-hand side from the field. Please refer to point 2.2.2.6 as this functionality is also included in the Scope tab

Within this window user is also able to adjust the Project duration for the whole Estimate. **Important**: change of this parameter has impact not only on the resource canvas but on the whole estimate.

- **All Default** – changes all values of the Start and End Weeks to the default ones
- **Save and Close** - saves latest values without displaying (compare with 'Assign') them to the user and closes the window
- **Assign** – saves latest values and displays Phases two rows above each week in Resources Canvas based on the schedule. If there are more than 2 Phases within one week there will be displayed only the first one with continuation sign.

When user decides to change Project duration while editing phases schedule, values will adjust accordingly to below rules:

- when start week and end week of given Phase are default (not changed by user), then after project duration change, they will adjust automatically,
- when either start week or end week are customized by the user, then the whole row (Phase) is treated as custom. That means that, after project duration change, they both will stay unchanged,

There is an exception to that rule, when values in the customized row exceed new Project duration. In that case, values are modified from custom to default (adjusting to the new project's period of time).

3.3.2.6 Dst Export

Our effortless DST tool has been designed for your convenience and has undergone rigorous testing. For your personalized use and verification, we have outlined a few simple steps for you to follow. You'll find the file to test under the 'Add a team with GDS resources' section.

DST Export steps:

1. Add a team with GDS resources: Start by creating a new team, ensuring it only includes GDS resources.

2. Apply FTEs/ Hours: Next, assign the desired full-time work hours for each resource in your team.

3.Expand grouping: Expand the group to view more specific details about the team members and their tasks. Click on the arrow next to the team name to achieve this.

4. Select competencies, sub-competencies, and skills: Every resource has tailored skills and competencies. Select these from the respective dropdown lists. (Remember to deselect the skill if you change the chosen competency or sub-competency afterward.)

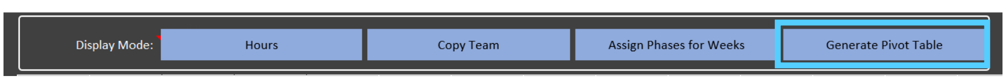
5. Export DST: Finally, by clicking the "Export DST" button, execute the function to export the competencies. The output will show separate workbooks for each competency, entirely comprising GDS resources.

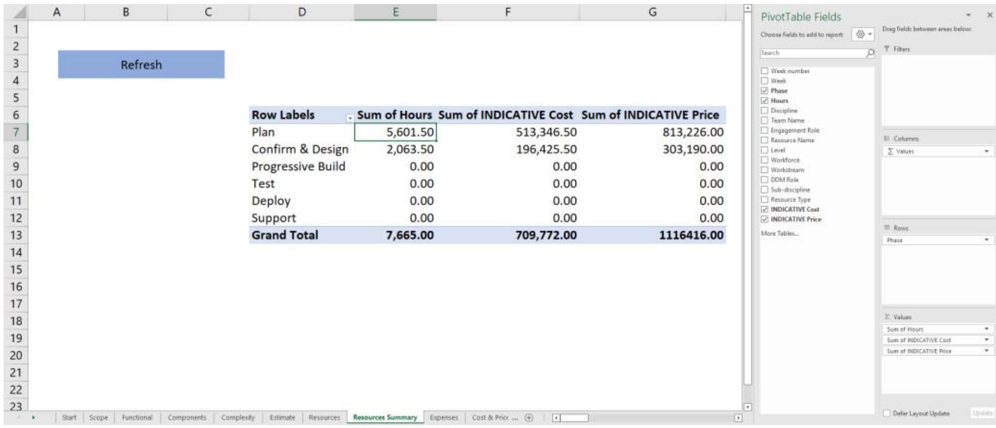
Note: If the project start date is in the past, you'll need to update it in the "Scope" tab.

Commented [AM13]: @Jakub Kątski@Mark Gurevich@David Falcone

Commented [DF14R13]: Accepted

3.3.2.7 Generate Pivot Table





Generate Pivot Table button creates a pivot table based on all data from Resources Canvas including newly assigned Phases per Week. This is a helpful tool for many types of summaries that user can expect. After clicking to generate pivot table, a new worksheet 'Resources Summary' will appear.

When clicking inside the table, a setting panel will appear on which user is able to choose parameters for the summary. It can be done by dragging a field name from window 'Choose fields to add to report' and dropping it to one of remaining windows: 'Filters', 'Columns', 'Rows', 'Values'.

It is necessary to assign Phases/Stages of the project to each week before generating the pivot table.

- **Refresh** button – updates the pivot table with the most current data.

3.3.2.8 Indicative financials

Indicative Financials

Total Billings	Margin	Blended Rate	GDS Ratio	Billings (Resources)	Cost (Resources)	Total Cost	EAF
$3,693,261.0	48.7%	$213.48	10.1%	$3,693,261.0	$1,893,061.3	$1,893,061.3	-2.6%

A dashboard of key financial indicators for the resource plan that is being constructed. The emphasis is on "indicative" as users still need to go through Mercury for the user numbers in the SOW. This section is explained more extensively in the Financials worksheet – section *3.6.2*:

- **Total billings** — items billed to the client, including billable expenses
- **Margin** — ratio between total margin and billings [Margin $/Billings]
- **Blended Rate** – ratio between billings and total staffed hours [Billings/Staffed Hours]
- **GDS Ratio** – ratio between GDS staffed hours and total staffed hours [GDS staffed hours/Total staffed hours]
- **Billings (resources)** — total billings, excluding billable expenses, currently included in the resource plan
- **Cost (resources)** — the total cost, excluding non-billable expenses of resources, currently included in the resource plan

- **Total cost** — includes both resources and non-billable expenses
- **EAF** — engagement adjustment factor [(Billings/NSR)-1]; typically, a negative number representing the client's discount rate as a percentage

3.3.2.9 Current Currency Selections

Current Curency Selections	
Country Currency	USA SDC
FX Code	USD
Symbol	$
1 USD =	1.0000
1 USD =	1.0000

- **Country Currency** – country of currency
- **FX Code** – foreign exchange global code for that country
- **Symbol** – symbol of the currency
- **1 USD =** - how much is 1 USD equivalent to the currency USD is being converted into
- **1 GDP =** - how much is 1 GDP of that country equivalent to USD

3.3.3 Using the Resources worksheet

There are five major usage scenarios to consider:

1. Single team semiautomated approach
2. Multiple team semiautomated approach
3. Manual approach
4. Working with multiple release
5. Working with long-running projects

3.3.3.1 Single team semiautomated approach

In this scenario, the user will leverage a pre-configured team setup to create an initial resource plan and then make manual adjustments to fine-tune the resulting plan. Steps for carrying out this scenario are as follows:

1. Set up a team of resources with the help of the *Add Configured Teams* function, invokable with a press of button. Users update the team name and start and end weeks as applicable within the context of the deal they are processing.
2. Plot FTEs/hours across the duration of the project by applying the *Apply FTEs/Hours* function. The FTEs or hours will be filled based on the team configuration parameters.
3. Review FTEs/hours and adjust the numbers to account for project ramp-up and ramp-down.
4. Review and compare estimated and staffed hours. Initially, users compare the numbers at the total level and then examine the breakdowns into workstreams or disciplines by expanding the section at the top of the Resources worksheet – section *3.3.2*.
5. Extend or shrink duration of individual teams and roles to gain the balance between the staffed and estimated hours. If necessary, users remove or add new roles.

6. To add a role (Developer 2, 3, etc.):
    a. Insert a blank row using the Insert Row function.
    b. Copy corresponding role that is closest in nature to what is required.
    c. Change role parameters such as name, level, etc.
7. Re-execute step 2 in order to update the staffed hour numbers.
8. Iterate steps 3 through 7 until estimated and staffed hours are roughly balanced.
9. Finally, review the Indicative Financials dashboard. Users update levels, workforce and other individual role parameters to achieve the desired financial metrics.

3.3.3.2 Multiple teams semiautomated approach

This is very similar to the single-team approach except that the user can select a different team for each major workstream of the project. For example, there might be multiple agile teams, each associated with a different application or application area of the project. The only difference here is that the process described in the previous section would have to be processed for each team.

3.3.3.3 Manual approach

This follows a slightly different process:

1. Create a new team by applying the Add New Team function. This will enable the drop-down in the Team Name column, enabling the user to add roles to the team.
2. Go through and add desired roles to the team by traversing the worksheet from left to right, row by row, using the drop-down or manual fill to enter role parameters.
3. Once all desired roles are filled, follow steps 2–9 from the single-team semiautomated approach.

3.3.3.4 Working with multiple releases

When working with project involving multiple release, consider the following steps:

1. First, create a resource plan that covers one release but plot the resource for the entire project.
2. Split off a section corresponding to release 2-N. Use *Copy-and-Split* function to achieve this. The split week is the last week of release 1.
3. Repeat the process until the resource plan is split into the desired number of releases.
4. For each release boundary, adjust the account for resources that must stay beyond the finishing date of the release and the resources that must start early and manually adjust FTEs/hours to accommodate that.
5. Also, modify individual releases for the differences in required resources such as moving some roles to GDS, updating resource levels, etc.

3.3.3.5 Working with long-running projects

A long-running project may cross over one or more fiscal years. In this scenario, consider the impact of cost rate increases and promotions:

1. Follow the process similar to the one described under "Working with multiple releases."
2. When invoking *Copy and Split Team*, select the split week that is on a boundary between fiscal years. Also, select the appropriate financial year to enable inclusion of correct cost rates.
3. Repeat the process if the project extends for more than two financial years.
4. In the section corresponding to each financial year, consider adjusting for resources that might be promoted. Alternatively, consider onboarding time for the resources who need to be

rolled on in place of resources that are rolled off. If the specific names of resources are not known, make best estimates based on experience regarding the number of resources that can be promoted.

3.3.4 Team Configurations

On the Team Config worksheet, the user will have access to pre-configured teams' structure, including resource's role, level, resource type, discipline and subdiscipline, GDS (Global Delivery Services) status, and default FTE percentage.

Overview		Import Team	Export Team									
Default Workstrear	Practice	Conf Team Nam	DDM Role Abb	DDM Role	Engagement Rol	Default Discipline	Sub-discipline	Resource Typ	Level	Workforc	Ramp Up (in Weeks)	Ramp Down (in Weeks
	DDM	Cross_dsp_lg	SQAE	Service Quality Assurance Executive	SQAE	Technical	Software	Client-serving	PP-7	Non-GDS		
	DDM	Cross_dsp_lg	DL	Delivery Lead	DL	Programmatic	NA	Client-serving	PP	Non-GDS		
	DDM	Cross_dsp_lg	FL	Func Lead	FL	Functional	NA	Client-serving	SM1	Non-GDS		
	DDM	Cross_dsp_lg	LA	Lead Architect	LA	Technical	NA	Client-serving	SM1	Non-GDS		
	DDM	Cross_dsp_lg	PM	Project Mgr	PM	Programmatic	NA	Client-serving	Mgr	Non-GDS		

Within this worksheet user can also see all pre-configured teams list as well as list of practices available in the model.

If there are no teams that meet user's requirements, there is a possibility to import a custom pre-configured team from dedicated user's template. The "Import Team" button is located above the main table in Team Config worksheet. After importing, team can be added automatically in the resource canvas.

To reuse custom-built team in the future in different estimates, user can Export pre-configured team into the template using "Export Team" button in Team Config worksheet. To be able to do this, firstly, the custom team should be saved using such Save Team to Config functionality in Resource worksheet.

3.4 Update financial parameters

3.4.1 Overview

There are two types of parameters to consider:

1. Cost & price rates
2. Expenses

In many scenarios, this step can be performed ahead of any other steps in the context of resource planning. Consider performing it early in the process.

3.4.1.1 Cost & price rates

The Estimator is populated with the cost and NSR rates that were initially loaded into Mercury at the point of time immediately prior to the tool's release. These rates are refreshed quarterly and can be imported into Estimator, allowing the user to apply the latest rates to obtain accurate indicative financials for estimates and resource plan being created.

The cost and NSR rates are installed and maintained for the current and three additional financial years.

In addition to cost and NSR rates, the Estimator allows importing of the rate card information. In order to import rate card information, users should apply the standard template that is included with the tool. By default, the rate information is not included in the Estimator and should either be imported or keyed in manually.

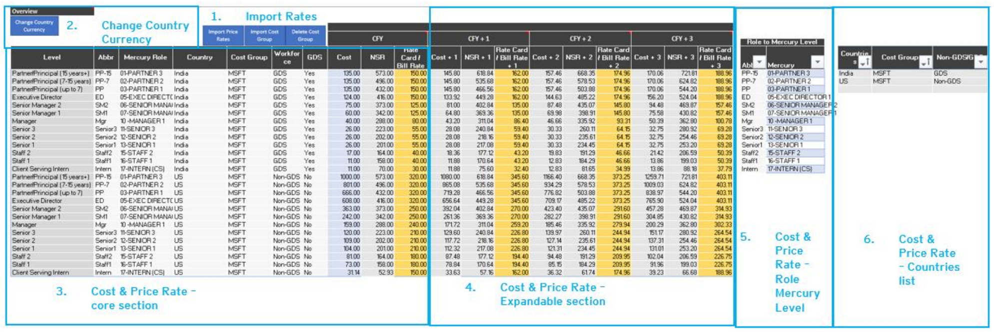
The rates processing is supported by the Cost & Price worksheet.

3.4.1.2 Expenses

A rudimentary expense tracking capability is provided with Estimator. Ability to identify type and category (e.g., billable vs. non-billable expenses) is provided. Summaries of expenses by categories are shown and are included in computations.

The processing of expenses is supported by the Expenses worksheet.

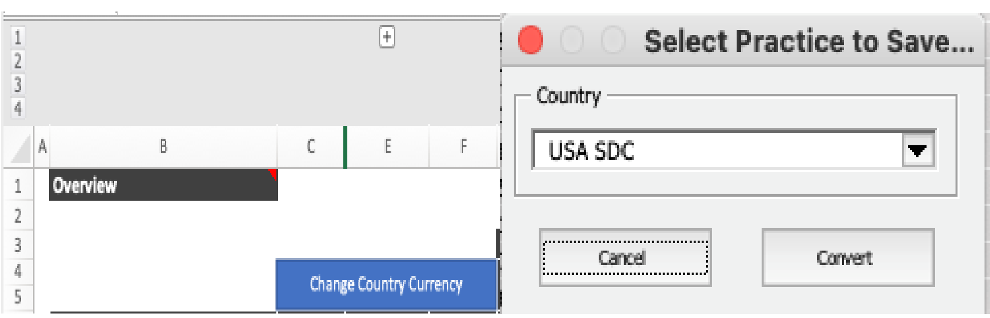
3.4.2 Cost & Price worksheet details



This worksheet emphasizes different rates for different level of resources based on specific fiscal year:

- **Change Country Currency –** allows users change currency value to another country
- **Cost & Price Rate — core section —** covers all levels of resources, their GDS status, cost, NSR and rate card values for current fiscal year
- **Cost & Price Rate — expandable section** —covers all levels of resources, their GDS status, cost, NSR and rate card values for the next three fiscal years
- **Role to Mercury Level —** abbreviated roles compared with roles in Mercury
- **Countries list –** lists the countries, cost groups, and if they are or are not GDS
- **Import rates –** forms that allow import of price rates and cost groups, and deletion of cost groups

3.4.2.1 Change Country Currency



By clicking on this button, user is prompted to select the country to which they would like to convert the currency to. Then a wizard appears where they would select that country and click on convert and then prices in the tables below would change accordingly.

3.4.2.2 Cost & Price Rate – core section

Change Country Currency

Import Price Rates | Import Cost Group | Delete Cost Group

Level	Abbr	Mercury Role	Country	Cost Group	Workforce	GDS	CFY Cost	CFY NSR	CFY Rate Card / Bill Rate
Partner/Principal (15 years+)	PP-15	01-PARTNER 3	India	MSFT	GDS	Yes	135.00	573.00	150.00
Partner/Principal (7-15 years)	PP-7	02-PARTNER 2	India	MSFT	GDS	Yes	135.00	496.00	150.00
Partner/Principal (up to 7)	PP	03-PARTNER 1	India	MSFT	GDS	Yes	135.00	432.00	150.00
Executive Director	ED	05-EXEC DIRECTO	India	MSFT	GDS	Yes	124.00	416.00	150.00
Senior Manager 2	SM2	06-SENIOR MANA	India	MSFT	GDS	Yes	75.00	373.00	125.00
Senior Manager 1	SM1	07-SENIOR MANA	India	MSFT	GDS	Yes	60.00	342.00	125.00
Manager	Mgr	10 -MANAGER 1	India	MSFT	GDS	Yes	40.00	288.00	80.00
Senior 3	Senior3	11-SENIOR 3	India	MSFT	GDS	Yes	26.00	223.00	55.00
Senior 2	Senior2	12-SENIOR 2	India	MSFT	GDS	Yes	26.00	202.00	55.00
Senior 1	Senior1	13-SENIOR 1	India	MSFT	GDS	Yes	26.00	201.00	55.00
Staff 2	Staff2	15-STAFF 2	India	MSFT	GDS	Yes	17.00	164.00	40.00
Staff 1	Staff1	16-STAFF 1	India	MSFT	GDS	Yes	11.00	158.00	40.00
Client Serving Intern	Intern	17-INTERN (CS)	India	MSFT	GDS	Yes	11.00	70.00	30.00
Partner/Principal (15 years+)	PP-15	01-PARTNER 3	US	MSFT	Non-GDS	No	1000.00	573.00	320.00
Partner/Principal (7-15 years)	PP-7	02-PARTNER 2	US	MSFT	Non-GDS	No	801.00	496.00	320.00
Partner/Principal (up to 7)	PP	03-PARTNER 1	US	MSFT	Non-GDS	No	666.00	432.00	320.00
Executive Director	ED	05-EXEC DIRECTO	US	MSFT	Non-GDS	No	608.00	416.00	320.00
Senior Manager 2	SM2	06-SENIOR MANA	US	MSFT	Non-GDS	No	363.00	373.00	250.00
Senior Manager 1	SM1	07-SENIOR MANA	US	MSFT	Non-GDS	No	242.00	342.00	250.00
Manager	Mgr	10 -MANAGER 1	US	MSFT	Non-GDS	No	159.00	288.00	240.00
Senior 3	Senior3	11-SENIOR 3	US	MSFT	Non-GDS	No	120.00	223.00	210.00
Senior 2	Senior2	12-SENIOR 2	US	MSFT	Non-GDS	No	109.00	202.00	210.00
Senior 1	Senior1	13-SENIOR 1	US	MSFT	Non-GDS	No	104.00	201.00	210.00
Staff 2	Staff2	15-STAFF 2	US	MSFT	Non-GDS	No	81.00	164.00	180.00
Staff 1	Staff1	16-STAFF 1	US	MSFT	Non-GDS	No	73.00	158.00	180.00
Client Serving Intern	Intern	17-INTERN (CS)	US	MSFT	Non-GDS	No	31.14	52.93	150.00

- **Level** – resource's EY level
- **Abbreviation** – resource level abbreviation
- **Mercury Role** - resource level in Mercury
- **Country** - resource location
- **Cost group** - cost center the resource is associated with
- **Workforce** - Global Delivery Service status for each resource (GDS or Non-GDS)
- **GDS** – Global Delivery Service status for each resource (Yes/No)
- **Cost** – rates for current fiscal year by resource level and GDS status
- **NSR** – net service revenue for current fiscal year by resource level and GDS status
- **Rate card/bill rate** – represents the rate to be presented to the client:
- Typically, equal to the NSR, bill rate can be overridden based on agreement with the client or specific agreement for a given engagement.

3.4.2.3 *Cost & Price Rate – expandable section*

CFY			CFY + 1			CFY + 2			CFY + 3		
Cost	NSR	Rate Card / Bill Rate	Cost + 1	NSR + 1	Rate Card / Bill Rate + 1	Cost + 2	NSR + 2	Rate Card / Bill Rate + 2	Cost + 3	NSR + 3	Rate Card / Bill Rate + 3
135.00	573.00	150.00	145.80	618.84	162.00	157.46	668.35	174.96	170.06	721.81	188.96
135.00	496.00	150.00	145.80	535.68	162.00	157.46	578.53	174.96	170.06	624.82	188.96
135.00	432.00	150.00	145.80	466.56	162.00	157.46	503.88	174.96	170.06	544.20	188.96
124.00	416.00	150.00	133.92	449.28	162.00	144.63	485.22	174.96	156.20	524.04	188.96
75.00	373.00	125.00	81.00	402.84	135.00	87.48	435.07	145.80	94.48	469.87	157.46
60.00	342.00	125.00	64.80	369.36	135.00	69.98	398.91	145.80	75.58	430.82	157.46
40.00	288.00	80.00	43.20	311.04	86.40	46.66	335.92	93.31	50.39	362.80	100.78
26.00	223.00	55.00	28.08	240.84	59.40	30.33	260.11	64.15	32.75	280.92	69.28
26.00	202.00	55.00	28.08	218.16	59.40	30.33	235.61	64.15	32.75	254.46	69.28
26.00	201.00	55.00	28.08	217.08	59.40	30.33	234.45	64.15	32.75	253.20	69.28
17.00	164.00	40.00	18.36	177.12	43.20	19.83	191.29	46.66	21.42	206.59	50.39
11.00	158.00	40.00	11.88	170.64	43.20	12.83	184.29	46.66	13.86	199.03	50.39
11.00	70.00	30.00	11.88	75.60	32.40	12.83	81.65	34.99	13.86	88.18	37.79

This covers all above-mentioned fields/columns for the next three fiscal years (CFY+1, CFY+2 and CFY+3).

3.4.2.4 *Role to Mercury Level*

Role to Mercury Level	
Abbr	Mercury
PP-15	01-PARTNER 3
PP-7	02-PARTNER 2
PP	03-PARTNER 1
ED	05-EXEC DIRECTOR 1
SM2	06-SENIOR MANAGER 2
SM1	07-SENIOR MANAGER 1
Mgr	10 -MANAGER 1
Senior3	11-SENIOR 3
Senior2	12-SENIOR 2
Senior1	13-SENIOR 1
Staff2	15-STAFF 2
Staff1	16-STAFF 1
Intern	

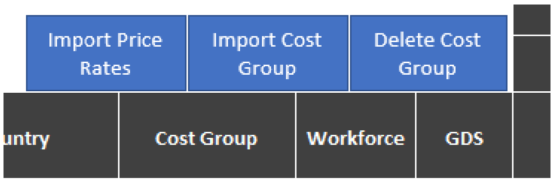
- **Abbr** – abbreviated name of engagement roles/levels mentioned in the workbook
- **Mercury** – their respective roles/levels in Mercury

3.4.2.5 *Countries List*

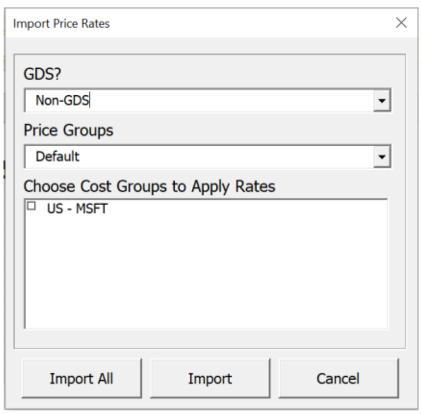
Countries	Cost Groups	Non-GDS/GDS
India	MSFT	GDS
US	MSFT	Non-GDS

- Lists the cost groups with their associated countries and GDS values that are populated in the Cost and Price main table. New cost groups will be added here when using the "Import Cost Group" form.

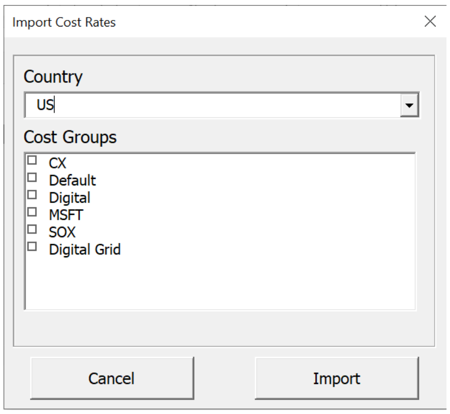
3.4.2.6 Countries List



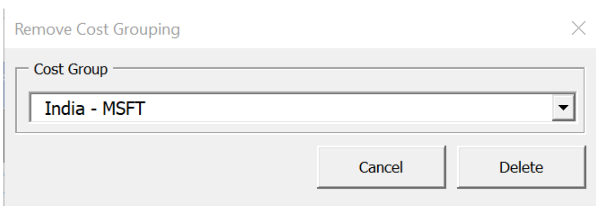
- **Import Price Rates –** Used to update the bill rates
  - Select the rates file you want to import. Once the rates file is selected, a form will appear which shows the available Cost Groups to apply rates to import



  -
  - Select either GDS or Non-GDS, choose a Price Group, select the cost groups to apply rates to (or click “Import All” to apply rates to all cost groups), and then click “Import”. Note - If “Import All” is selected, rates will update for either all GDS or all Non-GDS rates (based on selection)
  - Updated rates will appear in the “Bill Rate/ Rate card” column
- Import Cost Group –
  - Select the rates file you want to import. Once the rates file is selected, a form will appear which shows the available Cost Groups to import

- 

- Select the Country and Cost Groups, which load based on the country selected, and then click Import
- The Cost and Price table and Country table will then be updated with the selected cost groups

- **Delete Cost Group –** delete the cost group from the Cost and Price table
  - 

  - Select the Cost Group to delete from the Cost and Price table and click “Delete” to delete that group from the Cost and Price table as well as the Country table

3.4.3 Using the Cost & Price worksheet

There are two separate steps to consider:

3. Cost and NSR rates
4. Rate card

The process is similar though not identical for both.

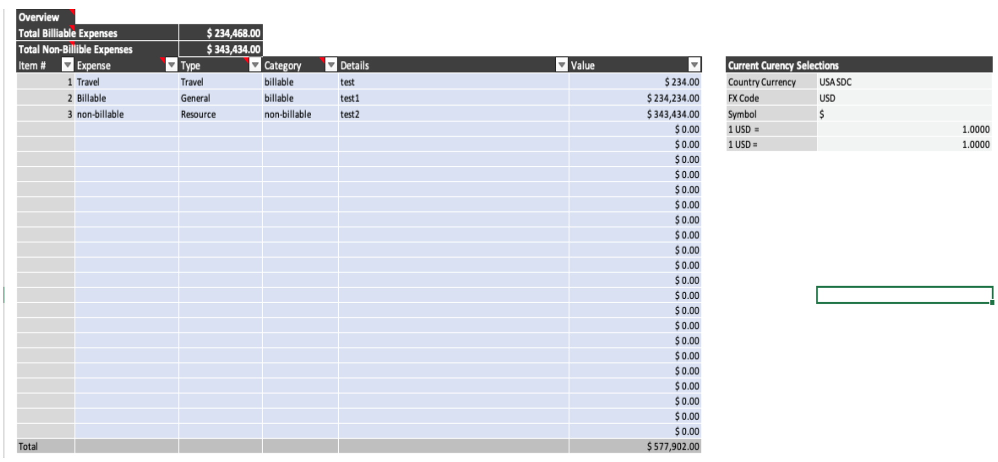
3.4.3.1 Cost and NSR rates

1. Review the last imported date.
2. If it is more than three months old, the recommendation is to obtain the latest rates file and re-import it:
   a. Download file from the Quarterly Rate File Location
   b. Click on “Import Cost & NSR Rates”
   c. Browse your machine and find the file that has been imported
   d. Open the file

3.4.3.2 Rate card

1. If the rate card rates have never been imported, then follow these steps:
    a. Click on "Import Rate Card Info"
    b. Override any rates established by the client master agreement or negotiated otherwise
2. If the rate card rate has already been imported at least once, then either:
    a. The rates can be refreshed by re-importing the rate card template.
    b. The user can override the individual rates that change directly in the Cost and Price worksheet in the Rate Card/Bill Rate column

3.4.4 Expenses worksheet details



Overview	
Total Billiable Expenses	$ 234,468.00
Total Non-Billible Expenses	$ 343,434.00

Item #	Expense	Type	Category	Details	Value
1	Travel	Travel	billable	test	$ 234.00
2	Billable	General	billable	test1	$ 234,234.00
3	non-billable	Resource	non-billable	test2	$ 343,434.00
					$ 0.00
					$ 0.00
					$ 0.00
					$ 0.00
					$ 0.00
					$ 0.00
					$ 0.00
					$ 0.00
					$ 0.00
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					$ 0.00
					$ 0.00
					$ 0.00
					$ 0.00
					$ 0.00
					$ 0.00
					$ 0.00
Total					$ 577,902.00

Current Curency Selections	
Country Currency	USA SDC
FX Code	USD
Symbol	$
1 USD =	1.0000
1 USD =	1.0000

This worksheet provides a basic capability for expense accounting. It can be used to enter a single item of total expenses by billable/non-billable categories or to create a detailed inventory that is calculated and then used for financial reporting:

- **Total billable expenses** — total pass-through expenses that are directly billable to the client (sum of all expenses that are categorized *billable*)
- **Total non-billable expenses** — total non-billable expenses that are added to the cost figure (sum of all expenses that are categorized *non-billable*)
- **Item number** — expense list ID
- **Expense** — a name given to the expense
- **Type** — expense category such as travel, general or resource (e.g., contractors)
- **Category** — specifies whether its billable or non-billable
- **Details** — brief explanation of the expense
- **Value** — dollar value of individual expense

Current Currency Selections

- **Country Currency** – country of currency

- **FX Code** – foreign exchange global code for that country
- **Symbol** – symbol of the currency
- **1 USD =** - how much is 1 USD equivalent to the currency USD is being converted into
- **1 GDP =** - how much is 1 GDP of that country equivalent to USD

3.4.5 Using the Expenses worksheet

Itemize expenses and enter them into the table on the Expenses worksheet. Identify the expense, then specify expense description, type, category and the amount. Review the total billable and non-billable expense at the top of the worksheet.

3.5 Refine and make adjustments

In this step, the user is encouraged to consider adjustments that can be applied to the resource plan that may improve deal economics as well as to review the risks that may undermine the delivery. There are several areas to consider, but key items include:

- Shifting work to GDS, especially in the multiple-release scenario (optimization)
- Increasing leverage of junior resources, especially on longer-running projects (optimization)
- Confirming skills availability, especially on engagements involving cutting-edge technologies (risk prevention)
- Reviewing the impact of COLA (cost-of-living adjustments) and resource raises and promotion, particularly on longer-running engagements
- Review indicative financials and confirm that goals for the pursuit are achievable
- Review balance between estimated and staffed hours

Shifting work to GDS — Consider shifting work to GDS or other lower-cost locations, taking into account skill availability, critical mass of functionality that can be delegated to other location(s), balance against the overall extra-communication and challenges with multiple time zone separation. On the multiple-release engagements, increase shift to GDS with each release. As the resources are shifted to GDS, review and fine-tune the estimate using Geographic distribution adjustments that are enabled on the Scope worksheet. This will help to properly account for the overhead of working in multiple locations.

Increasing leverage of more junior resources — Start with senior resources, especially in the roles of architects and functional leads. On the longer-running engagements, consider bringing more junior resources into these roles. For example, have a manager-level architect start on the engagement. After a few months, include a junior resource, e.g., senior 3, in that role for one to three months, and then either reduce the time or consider rolling off the senior resource.

Confirm skill availability — Verify that the required skilled resources are available when needed. Adjust the resource plan and prepare contingency plans in case they are not, e.g., consider using contractors and update the resource plan accordingly.

Review impact of COLA and resource promotions — For longer-running engagements that may span multiple fiscal years, take into account COLA impacts on cost rates. Use the Estimator’s features to bring appropriate rates that should switch in the resources on the fiscal year boundaries. Also, consider impact of resource promotion and the options for handling it. One option is to roll off the resources that are promoted and bring on junior resource — be sure to include overlaps between resources to account for onboarding and knowledge transfer. Alternatively, keep resources as they are accounted for it in the resource plan and evaluate impact on the margin.

Keep an eye on estimated vs. staffed hours — As updates to the resource plan are made, watch the balance between estimated and staffed hours for each category (discipline or workstream). Add or remove resources and adjust hours/FTEs for each resource to align each category (+/-5%).

Review indicative financials — Confirm that margin and EAF requirements are met for the deal in pursuit.

Key features to consider while making adjustments for the items described above include:

- To add a new resource, use *Insert Row* function to add new resources and then manually enter resource parameters
- To adjust estimate for geographic distribution, make changes on the Scope worksheet, then add new resources/extend time to compensate for the increase effort estimate
- To switch resources from EY staff to contractors, adjust the individual resource parameters and add contractors as cost on the Expenses worksheet
- Use *Copy and Split* function to address multiple releases and long-running engagements; change fiscal year when applying this function
- Review Indicative Financial dashboard on the Resources worksheet, along with detailed financial and other key ratio metrics on the Financials worksheet
- Review estimated vs. staffed hours in expandable section at the top of the Resources worksheet or a separate table on the Financials worksheet

3.6 Review indicative financials

3.6.1 Overview

The purpose of this step is to help meet the financial targets for the pursuit effort. Several aspects should be monitored:

- **Indicative financial metrics** — standard financial parameters such as cost, price, margin, etc.
- **Contingency** — inclusion of contingency and the matter in which it is handled in both the estimate and the resource plan
- **Staffing pyramids and ratios** — percent distribution of staff hours by level
- **Distribution of effort across different workforces**

The provided metrics and ratios allow adherence to the established financial targets and validation of these targets as part of the SQAE (Service Quality Assurance Executive) review process.

Note that initial review can be accomplished by looking at the Indicative Financial dashboard on the Resources worksheet. The dashboard allows monitoring of key metrics in flight, as the resource plan is modified. Additional metrics are available on the Financials worksheet, which is detailed below.

3.6.2 Financials worksheet details

This worksheet summarizes the financial metrics of a resource plan. It is primarily broken down into the following parts:

- **Indicative financials** – contains overall financial computations for the resource plan engine
- **Ratios** – key ratios based on resource level/roles
- **Current Currency Selections** - present country's currency selections
- **Save Final Estimate** – saves the latest version of the estimate for the user
- **Estimated vs. staffed hours by discipline** – breaks down estimated versus staffed hours by discipline stage
- **Staffing pyramid** – breaks down staffed hours by resource level
- **Workforce effort** – breaks down effort distribution by resource type
- **Total hours** – total number of estimated hours and staffed hours

3.6.2.1 Indicative financials

Indicative Financials	
Total Cost	$1,845,391.9
Cost	$1,844,614.9
Non-billable Expenses	$777.0
NSR	$3,661,402.4
Total Billings	$2,871,038.1
Billings	$2,869,905.1
Billable Exps	$1,133.0
Margin $	$1,024,513.3
Margin %	35.7%
Blended Rate	$170.5
Travel Expense %	0.0%
EAF	-21.6%

- **Total cost** – sum of cost and non-billable expenses:
  - **Cost** – all indicative cost values for all teams added under Resources worksheet
  - **Non-billable expenses** – all expenses under the category *Non-billable* under Expenses worksheet
- **NSR** – sum of all net service revenue values for all teams added under Resources worksheet
- **Total billings** – sum of billings and billable expenses:
  - **Billings** – all indicative price values for all the teams added under Resources worksheet
  - **Billable expenses** – all expenses under the category *Billable* under Expenses worksheet
- **Margin cost** – total dollar value of margin is the difference between billings and cost value
- **Margin percentage** – the ratio between $ and Billings
- **Blended Rate** - blended rate value is equals to Billings divided by the total hours assigned to resources for the duration of an engagement
- **Travel Expense %** - Billable Expenses / Total billings
- **EAF** – the percentage discount rate for an engagement, calculated as (Billings/NSR)-1

3.6.2.2 Ratios

Key Ratios	
PPMD Ratio	0.0%
Staff Ratio	0.0%
GDS Ratio	0.0%

Expense Ratio	
	0.0000%

- **Key ratios:**
  - **PPED ratio** – percentage of PPMD total hours relative to total resource hours

    - **Staff ratio** – percentage of staff total hours relative to total resource hours
    - **GDS ratio** – percentage of GDS total hours relative to total resource hours
- **Expense ratio** – Calculates the ratio of total billable expenses to total resource billings

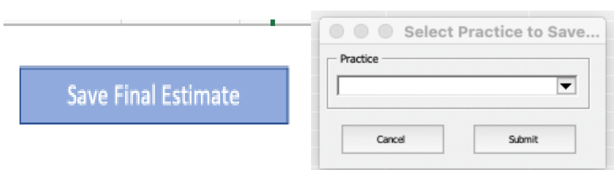
3.6.2.3 Current Country Selections

Current Curency Selections	
Country Currency	USA SDC
FX Code	USD
Symbol	$
1 USD =	1.0000
1 USD =	1.0000

Current Currency Selections

- **Country Currency** – country of currency
- **FX Code** – foreign exchange global code for that country
- **Symbol** – symbol of the currency
- **1 USD =** - how much is 1 USD equivalent to the currency USD is being converted into
- **1 GDP =** - how much is 1 GDP of that country equivalent to USD

3.6.2.4 Save Final Estimate



This button allows user to save the latest version of the estimate that they are working on. Once user clicks on the button, it asks them to select a practice from which this estimator will be imported into and then get saved as a file. Each estimator will save stored into a folder called Completed Estimators in the Teams channel.

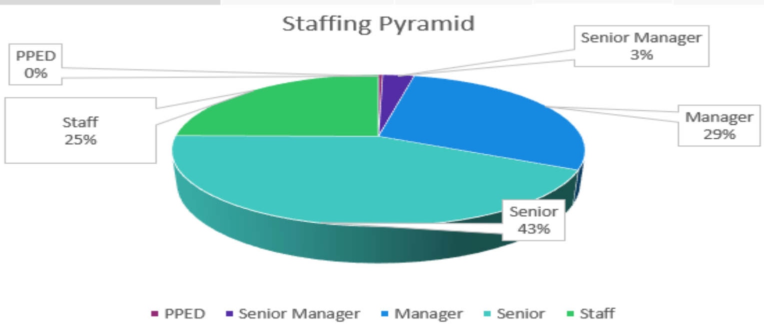
3.6.2.5 Estimated vs. staffed hours by discipline

Estimated vs. Staffed Hours by Discipline		
Discipline	Estimated	Staffed
Programmatic	0.0	1870.0
Functional	15983.8	54000.0
Technical	2244.0	13500.0
Change Mgt	310.0	0.0
UX/UI	0.0	0.0
Total	**18537.8**	**69370.0**

- **Discipline** – drop-down list of disciplines for the estimation
- **Estimated** – breakdown of estimated hours by discipline
- **Staffed** – breakdown of actual staffed hours by discipline

3.6.2.6 Staffing pyramid

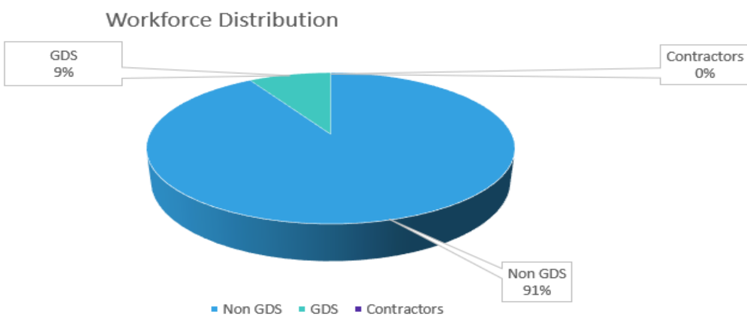
Staffing Pyramid Level	Hours		%	Hours
PPED			0.4%	66.2
Senior Manager			2.9%	496.0
Manager			28.7%	4826.3
Senior			43.1%	7256.3
Staff			24.9%	4185.0
TOTAL			100.0%	16829.7



- **Level** – EY level of resources in the engagement
- **Percentage** – allotment of staffed hours by resource level against the total staffed hours
- **Hours** – total staffed hours by each resource level

3.6.2.7 Workforce effort

Workforce Effort Distribution		
Type	%	Staffed Hours
Non GDS	91.4%	15378.4
GDS	8.6%	1451.3
Contractors	0.0%	0.0
TOTAL	100.0%	16829.7



- **Type** – category of resources in the engagement
- **Percentage** – allotment of staffed hours by resource type against the total staffed hours
- **Staffed hours** – total staffed hours by each resource type

3.6.2.8 Team parameters

Team Parameters				
Team Name	Workstream	Configured Tear	Start Week	End Week
MSERP		ERPSmall	1	26
MSERP2		ERPMedium	1	26
MSERP3		ERPLarge	1	13
DC			1	12
Test30			1	20

- **Team name** – name of the teams added in the Resources worksheet
- **Workstream** – field of activity for each team
- **Configured team** – the name of pre-configured team that this team includes
- **Start week** – beginning week of the engagement
- **End week** – last week of the engagement

3.6.2.9 Total hours

Total Hours	
Estimated	Staffed
8198.2	16829.7

- **Estimated** — total estimated hours calculated for all the resources
- **Staffed** — total staffed hours calculated for all the resources

3.6.3 Using the Financials worksheet

This worksheet is self-explanatory, and there is no particular order in which one should review the metrics and ratios. Two special aspects should be mentioned:

- **Contingency inclusion/exclusion** — There is a capability to include/exclude contingency when comparing estimated and staffed hours. If the estimated hours include the contingency, then the matching staffed hours will have to be increased to achieve the alignment.
- **Team table** — This table is normally closed in a collapsed section and is not intended for the end users. Team table tracks all teams included in the resource plan. It may be of interest only when there are many teams and the user want to see the start/end week settings for all teams.

4 Review and refine

By and large, the key topics highlighted in this step have already been covered. This step is intended to cover changes that come external to the proposal pursuit process. Three sets of actions are considered:

- **Changes requested by the client** — these may come as a result of either:
  - **A change to the specific need, causing a change in scope/requirements** — Typically, this will result in change to size driver-specific selection or count, ultimately resulting in the estimate change. The change in estimate must propagate into a change in resource plan.
  - **A change in the client's budgeting process** — This also causes a change in scope. As described in the bullet above, the client knows specifically what it must cut, and the proposal team simply needs to go through a due diligence process and reduce the estimate and staffing levels accordingly and provide a new price point. In contrast, in this case, the client might be looking for options, and the proposal team/solution architect needs to recommend what might be cut and include the corresponding impact on price.

  In either case, a change needs to be propagated with the impact on effort estimate, resource plan and the new price point.
- **Changes due to competitive pressures** — A team might be involved in an RFP that is being responded to by several competitive vendors. In such a circumstance, there is a strong incentive to provide an EY solution at the lowest price point that satisfies EY margin requirements. In such a case, several levers can be pulled, as described in section *3.5*, "Refine and make adjustments."
- **Reacting to skill availability** — Part of due diligence is to confirm availability of the skills embedded in the resource plan. The topic is also covered in section *3.5*, "Refine and make adjustments." The actual skills availability review is outside the scope of estimating or resource planning. However, the end result of the review must be reflected in both the estimate and the resource plan.

5 Special topics

5.1 Assumptions

This specific worksheet prints out any rationale/comments, assumptions, change in value for any parameters and size and complexity drivers. As stated earlier, in Start worksheet, user can click on Assumptions after making above changes and it will direct them to this new worksheet. User can also click on Update Assumptions button which will include any new changes and distribute 2 tables – one for Object Inventory table in Components worksheet and rest of the changes are captured in the table below.

Assumptions Summary Component Object Inventory

Assumptions Summary Component Object Inventory

ID	Type	Workstream	Components	Object Type	Description	Size	Mode	Count	Default	Override	Adjustor	Rationale
1	Object Spec & Adj	Business Process Design	Default	Business Process - Business Process Design	Description: Future state business processes that will be designed for the target system. Includes: business process current state assessment, business process inventory, future state process documentation, process design workshops. Small: Business process requires developing Visio process flows only. Medium: Business process requires full business process specification document. Large: Business process requires full business process specification document and requirements definition.	medium	new	5	203	304.5	150.00%	Test

i. **Type** – types of changes in this table (i.e., Object Spec suggests rationale changes and Adj means when user changes any adjustor values)
ii. **Workstream** – which workstream is this change coming from
iii. **Components** – which component is this change coming from
iv. **Object Type** – name of the object
v. **Description** – brief description if available
vi. **Size** – size of the object
vii. **Mode** – current mode of the object
viii. **Count** – count of the object
ix. **Default** – the default value of that item
x. **Override** – the overridden value from that object
xi. **Adjustor** – for any size drivers that are printed into this table
xii. **Rationale** – any rationale provided

Assumptions Summary

Assumptions Summary

ID	Type	Worksheet	Table	Name	Description	Default	Override	Adjustor	Count	Comment
1	Assumption	Start	Assumption Table							Assumption Example #2
2	Assumption	Start	Assumption Table							Assumption Example #1
3	Override	Complexity	Complexity Table	Number of OpCos	Number of operating companies (OpCos) the utility has	simple	complex			Criteria Description: Utility with more than three OpCos
4										

xiii. **Type** – types of changes in this table
xiv. **Worksheet** – which worksheet is this change coming from

xv. **Table** – which table is this change coming from
xvi. **Name** – name of the assumption, parameter, size and complexity drivers
xvii. **Description** – brief description if available
xviii. **Default** – the default value of that item
xix. **Override** – the overridden value from that table/worksheet
xx. **Adjustor** – for any size drivers that are printed into this table
xxi. **Count** – count for the size drivers
xxii. **Comment** – any rationale provided in the original worksheet

5.2 Starting with resource plan

On occasion, very infrequently, one may start using the Estimator by establishing a resource plan. In such a case, the tool is being used as a sophisticated worksheet and is not recommended. Such usage may be justified for establishing team structure for staff augmentation projects.

Such an approach is also quite common for assessment and strategy projects. In this case, a pre-configured assessment or strategy team is used to define an initial staffing plan, which is then manually adjusted to match the client requirements.

5.3 Aligning estimate and resource plan

A project estimate and a resource plan are structured to support a specific project for a domain/subdomain such as MSFT M360 ERP. While the estimate is established by entering a number size and complexity drivers, a resource plan typically has one to three configurations, such as small, medium and large team. The total effort captured in the resource plan must be roughly equal to that established in the estimate. When such approximate equality occurs (+/- 5%), it is said that the estimate and the resource plan are aligned. Furthermore, the alignment should occur not just at the overall level but a lower level of granularity such as DDM discipline (programmatic, technical, functional, UX (user experience), etc.).

The support for achieving alignment is provided on the Resources worksheet — an expandable section at the top of the worksheet as well as a separate table on the Financials worksheet.

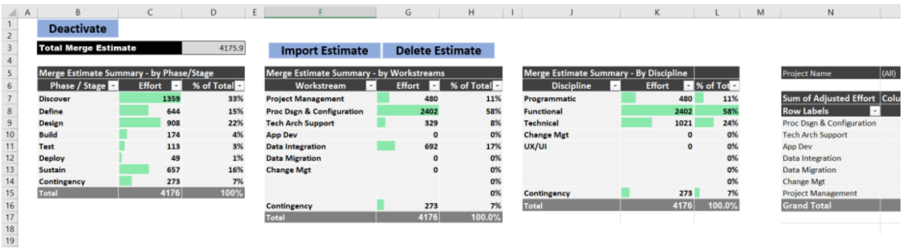
5.4 Refreshing cost and pricing rates

Obtaining accurate rates is critical to the success of the Estimator. These rates are exported from Mercury by a technology operations team member and are reviewed quarterly. They are stored in a secure team site and used to update the tool. Access to the rates mirrors the access users have in Mercury.

Users can load their own rates into the Estimator. This is done in cases where the client rate card is not used or for simulating rate changes. To add a rate to the estimator, the Estimator user can delete the rate for a resource and enter the new rate. Users can refer to section ***Error! Reference source not found.*** for further details.

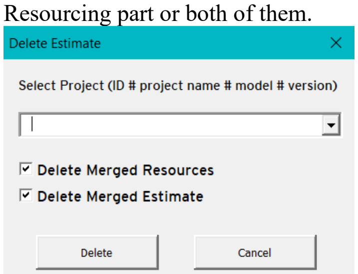
5.5 Merging Estimates

Merge Estimates feature is to provide user a detailed overview on all of the user's projects' estimates regardless of the model they use. In other words, user is able to see the summary of all projects in one workbook.



Buttons:

- Import Estimate – imports the project selected by the user
- Delete Estimate – opens a user form that enables to select one of the merged projects to be removed from current workbook. Additionally user can select whether to delete Estimate part or Resourcing part or both of them.



- Deactivate – **<u>removes all data from merged projects</u>** and hides the "Merge Summary" worksheet.

Feature is divided in two parts:

5.5.1 Estimation merge

New worksheet "Merge Summary" is shown to display a summary of combined estimation. Summary consists of 4 tables. Three of them are very similar to what we have in the Estimate tab, but containing data from all merged workbooks. Estimate tab will always refer to the current project. Details as follows:

a) Summary by phases/stages – this table displays total effort (from every merged estimates including current one) broken down into **current model's stages**. In several models phases' names differentiate from each other or their counts are unequal. In such cases, all models that are being merged, have their effort distribution converted into current phases.

Merge Estimate Summary - by Phase/Stage		
Phase / Stage	**Effort**	**% of Total**
Discover	1359	33%
Define	644	15%
Design	908	22%
Build	174	4%
Test	113	3%
Deploy	49	1%
Sustain	657	16%
Contingency	273	7%
Total	4176	100%

b) Summary by workstreams - this table displays total effort (from every merged estimates including current one) broken down into workstreams. Unlike phases, here are shown all workstreams present in all merged estimates. This means that if merged projects have common workstream names, their effort will be combined, else the workstream will be added as a new line to this table.

Merge Estimate Summary - by Workstreams		
Workstream	**Effort**	**% of Total**
Project Management	480	11%
Proc Dsgn & Configuration	2402	58%
Tech Arch Support	329	8%
App Dev	0	0%
Data Integration	692	17%
Data Migration	0	0%
Change Mgt	0	0%
		0%
		0%
Contingency	273	7%
Total	4176	100.0%

c) Summary by disciplines - this table displays total effort (from every merged estimates including current one) broken down into disciplines. **The table is shown only if all merged estimates support Disciplines**.

d) Summary pivot table – this is a standard pivot table holding data from all merged projects. User is able to compare data from different selected estimates or from one of them. The useful example would be verifying how the phases conversion went for a merged model that has different phases than the current one. In that scenario user needs to select only the merged project in the pivot table filter.

Project Name	(All)

Sum of Adjusted Effort	Column Labels							
Row Labels	**Discover**	**Define**	**Design**	**Build**	**Test**	**Deploy**	**Sustain**	**Grand Total**
Proc Dsgn & Configuration	847.1	520.7	591.3	0.0	0.0	0.0	442.9	2402.0
Tech Arch Support	115.9	28.4	70.8	43.1	6.6	3.2	60.6	328.6
App Dev	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Data Integration	261.1	30.7	84.9	100.0	86.1	41.0	88.7	692.5
Data Migration	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Change Mgt	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Project Management	134.6	63.8	160.6	30.8	19.9	4.9	65.1	479.7
Grand Total	1359	644	908	174	113	49	657	3903

This part is only summary. User cannot edit the part of merged estimation.

5.5.2 Resourcing merge

All resourcing data is being imported into current workbook including:

a) Resourcing teams in resources canvas – team names from merged project will get a label in brackets with the project's name. Example: merged project's name: ABC, Team name: Tech Arch Large; team name after merging: Tech Arch Large (ABC)
   While importing teams to resourcing canvas user is able to choose where on timeline to put them. Teams can be imported with
   i) no modification (original Week1 = current Week1),
   ii) with the shift of given number of weeks (example: original Week1 will become current Week 5)
   iii) respect of dates (team starting in Oct 1st will also start in Oct 1st regardless of Week number)
b) Cost Groups – if the cost group is already in current workbook, then user will be asked whether to skip importing this cost group or overwrite the existing one.
c) Pre-configured Teams – if pre-configured team is already in current workbook, then user will be asked whether to skip importing this team or overwrite the existing one
d) Expenses – expenses' names will also get project's name label.

Since all resourcing data is being imported, user is able to modify it.

5.6 Sharing model with the client

The Estimator represents EY intellectual property and needs to be guarded and treated with care. When a user needs to share estimate details with the client, share screenshots with the key estimator worksheets. Sharing an Estimator with a client may eventually lead to us losing the control of this valuable EY asset. In the future, a policy will be established on how an Estimator can be shared with clients, which may involve some form of monetary or other compensation, with strict limitations on how or whether the client can share an Estimator with other entities.

5.7 Setting key adjustors

Some key adjustors used for the estimation in this workbook are as follows:

- Fixed effort
- Scaler
- Override

5.7.1 Fixed effort

An estimate consists of fixed and variable effort components. As the name would imply, the fixed component does not vary (or varies only slightly very little) with the size and complexity of the project. The variable component is wholly dependent on the size and complexity of the project. The fixed component is represented by the fixed effort in the Estimator. It captures setup activities such as organizing meeting, conducting initial meetings with the pattern, setting up tools and environment, etc.

Within Estimator, fixed effort is represented with a fixed number of hours per stage per workstream. Thus, a base estimate for a workstream is equal to the summation of variable portion of the estimate for a given element, calculated based on size driver information, plus the fixed effort hours for that stage.

5.7.2 Scaler

Scaler adjusts the total effort by a scaler factor. It is used in a few estimating models (e.g., mobile), typically in applications that must run on multiple platforms and must accommodate multiple form factors. In terms of estimating total effort hours for a workstream/stage/discipline, scaler is used as a key numerical adjustor within the Object Inventory table. Currently, while this feature is supported, no model is using it yet.

Workstream	Scaler	Plan	Confirm & Design	Progressive Build	Test	Deploy	Support
Business Process Design & Config	Languages				2-4: 10% ; 5+: 5%		
	Business Unit		2-5: 10% ; 6-15: 6% ; 16+: 1.3%	2-5: 7% ; 6-15: 3% ; 16+: 1%	2-5 - 1% ; 6-10 - 0.5% ; 11+ - 0.2%		
	Ledgers		2-5: 10% ; 6-15: 5% ; 16+: 1.3%		2-5: 1% ; 6-10: 0.5% ; 11+: 0.2%		
	Months of Hypercare						2: 75% ; 3+: 50%
Enhancements	Languages		2-4: 5% ; 5+: 2%	2-4: 10% ; 5+: 5%	2-4: 10% ; 5+: 5%		
	SIT Cycles				2-4: 20% ; 5+: 10%		
	UAT Cycles				2-4: 20% ; 5+: 10%		2: 75% ; 3+: 50%
	Months of Hypercare						2: 75% ; 3+: 50%
Reports	Languages		2-4: 5% ; 5+: 2%	2-4: 10% ; 5+: 5%	2-4: 10% ; 5+: 5		
	SIT Cycles				2-4: 20% ; 5+: 10%		
	UAT Cycles				2-4: 20% ; 5+: 10%		
	Months of Hypercare						2: 75% ; 3+: 50%
Data Integration	SIT Cycles				2-4: 20% ; 5+: 10%		
	UAT Cycles				2-4: 20% ; 5+: 10%		
	Months of Hypercare						2: 75% ; 3+: 50%
Data Conversion	Migration cycles				2+: 80%		
	RACOR		-10%	-10%	-10%		
	Months of Hypercare						2: 75% ; 3+: 50%
	Languages		2-4: 5% ; 5+: 2%	2-4: 10% ; 5+: 5	2-4: 10% ; 5+: 5%		
Change Management	Months of Hypercare						2: 75% ; 3+: 50%
Technical Architecture	Application Landscape		1-5: 0% ; 6-15: 5% ; 16+: 2.5%	1-5: 0% ; 6-15: 5% ; 16+: 2.5%	1-5: 0% ; 6-15: 2.5% ; 16-25: 1.5%		
	Cloud Environments		1-4: 0% ; 5-9: 10% ; 10+: 5%	1-4: 0% ; 5-9: 10% ; 10+: 5%	1-4: 0% ; 5-9: 10% ; 10+: 5%		
	Integration Platforms		2: 20% ; 3-4: 15% ; 4+: 5%	2: 15% ; 3-4: 10% ; 4+: 5%	2: 10% ; 3+: 5%	2+: 5%	2+: 5%
	Months of Hypercare						2: 75% ; 3+: 50%
Security & Compliance	Months of Hypercare						2: 75% ; 3+: 50%

- **Workstream** – List of workstreams with scalers
- **Scaler** – Scaler name that affects the given workstream
- **Phase Columns (Plan/Confirm & Design/Progressive Build/Test/Deploy/Support)** – Lists each scaler's impact % on the phases of the given workstream
  - **Linear Scaler Impact** – Means the listed scaler impact % is applied to each additional unit (after 1) of the scaler
    - Denoted by 2+ followed by its scaler impact (i.e., 2+: 5%)
    - Ex – based off the table above and given 3 Migration Cycles, the Test Phase's efforts for Conversion would be calculated as:
      base effort * [1 + ((3-1) * 80%)]
  - **Linear with Breakpoints Scaler Impact** – Means there are different scaler impact %s applied to each of the listed ranges
    - Denoted by ranges followed by their scaler impact (i.e., 2-5: 7%; 6-15: 3%; 16+: 1%)
    - Ex – based off the table above and given 12 Business Units, the Test Phase's efforts for Business Process would be calculated as:
      base effort * [1 + ([(5-1) *1%] + [(11-6) *0.5%] + [((12-11) +1) *0.2%])]

5.7.3 Override

There are several parameters on the Scope and other worksheets that allow for overrides. These are typically provided a default value and the override in a separate column. This allows for the defaults to remain permanently visible, while override value can change.

For example, override values for both hypercare parameters, number of resources and number of weeks are used to calculate the total effort for hypercare. Similarly, geographic distribution parameters such as percentage of effort off-site and percentage adjustment for design, build, test and project management and contingency provide the ability for both default and override values.

The override parameters are also paired with a rationale column to allow for a quick identification of places where the default parameters were changed during a review process. The rationale provides clear explanation as to why the default parameter had to be overwritten. If the rationale is provided, then the override value will be highlighted yellow for the ease of noticing it. If no rationale is provided, then the override value will be highlighted red, which implies missing information.

5.8 Estimating discover phase

The discover phase can be estimated in one of the following ways:

- As small, medium and large fixed effort. To estimate it in this fashion, select "Effort Size" for the discover estimate type, then pick *small*, *medium* or *large* for the discover scale parameter.
- As a percentage of the rest of the phases of work, excluding project management. To estimate it using this method, simply select "% of rest of effort" for the discover estimate type parameter.

5.9 Operating in a basic mode

During client delivery, it is common to need to estimate a change request (CR). Typically, users need to estimate the effort for building one or more components. In such a scenario, the estimate needs to be completed without overhead such as project management, contingency, tech support, change management, etc. To accomplish this, switch the mode on the Scope worksheet.

Another example of using the basic mode is when a user is just starting out with the Estimator and wants to see its generic calculations and functionality. Following the overall flow of numbers is easier in basic mode.

5.10 Usage on Mac computers

The estimator is designed to work on PCs as well as Macs, with the functionality and the interface appearing similar to the end users on both. There are some exceptions. More specifically, the "whiting out" effect occurs during running of the following functions:

- **Apply FTEs/hours** — to apply either FTEs or hours for the team added in the resource plan
- **Display mode (FTE/hours)** — when trying to switch the display view between FTE or hours
- **Copy and split team** — trying to split a team into multiple teams by breaking down the work weeks
- **Save Final Estimator** – currently this functionality is not available on MAC, but the team is working towards a quick fix

5.11 Error handling

The estimator is equipped to handle errors during execution of any macros, especially in the Resources worksheet. When any of the macros such as Apply FTEs/Hours or Copy & Split Team is triggered, the Error Handling module will be invoked if there are any errors, i.e., calculation/ computation, formulas etc. Initially a folder would be auto generated in the C: drive named Estimator Project *(C:\Estimator Project)*. In that folder, a log file, in the format of *EstimatorLogFile-<current date>* would be automatically created if there is any error, containing details such as description, source, procedure name, location, worksheet name, line number, error Number and cell address. All errors occurring in the same date will be auto captured in the same date file whereas if any errors are captured in any following days, there would be a new log file for those respective dates.

5.12 Submitting enhancement requests or questions

Please email (estimators@ey.com) with suggested enhancements and/or questions. The team will assess the enhancement request and prioritize them in the product backlog in the backlog. Questions will be answered by the support team within 24 hours.

6 Appendix

6.1 Specialized topics

This section focuses on specialized topics and will be expanded over time as a result of the user's feedback:

- *How do I estimate super-large components? Objects?*
  There are a couple of options for doing this. First, isolate the individual objects that user consider to be super-large and use the line-item adjustor (provide reference to the right description of the column for the Components worksheet). The change 100% setting to 200% or 300% or to whatever user thinks is appropriate. Alternatively, instead of using count of 1 for a large size driver, increase unit count (again reference to the right place) to 2 or 3 or whatever is appropriate.
- *How do I accommodate super-simple complexity?*
  Use line-item adjustor and reduce 100% to 50% or lower.
- *Why does estimate jump so much upon entry of the first object on the Component worksheet?*
  The Estimator contains both variable and fixed effort estimation components. The variable part changes with the number of unit user enters for different drivers. The fixed effort, as the name implies, is fixed. Examples of fixed effort include tool configuration, technical architecture support activities, project management financial and status reporting per unit of time, etc. The Estimator is designed to show 0 effort estimate until at least one size driver is entered. Once the first size driver is entered, all the relevant fixed effort will be included — hence the immediate jump in effort hours.

6.2 Estimating change management

The description of change management is estimated. Two components — explicit object/component inventory and a small percentage of effort — are included in project management to account for initiative and leadership communication and other standard change management activities. This latter component of change management is referenced as "Change Management Communication" and can be included or excluded from scope.

6.3 Instant feedback approach

The Estimator has been designed to provide instant feedback on the impact of changes made by the end user. Any change in size driver type, mode, number unit selection, complexity selections, etc., is immediately reflected in the total estimate shown at the top of each worksheet and the summary section, which can be expanded. This is in contrast to the wizard-based or form-based approach that requires the user to enter all of the data, then press a button in order to observe the result.

6.4 Complexity vs. size concepts

Estimating is a two-dimensional problem, with one dimension being the size of the effort being estimated and the other the complexity. All four combinations are possible: (1) smaller size, simple complexity; (2) large size, simple complexity; (3) small size, significant complexity; and (4) efforts that are both large and complex. Hence, the estimating tool has both size and complexity drivers.

The notion can be demonstrated in the process of estimating the price of a house. If you ask a builder what the price of home will be, that person will ask you:

- What's the square footage of the house you want?
- Do you want basic, standard or a luxury house?

The first question addresses the notion of size, while the second one is equivalent to the notion of complexity.

It is the combination of both, that helps produce the home price estimate.

6.5 Financial summary computations

Major computations that must be mentioned to comprehend and use this workbook are as follow:

- **Indicative cost** = [CFY Cost Value (Cost and Price worksheet) * Number of Weeks (Resources worksheet) * Number of Default Work Hours (Resources worksheet)]
- **Indicative price** = [Pricing Rate Value (Cost and Price worksheet) * Number of Weeks (Resources worksheet) * Number of Default Work Hours (Resources worksheet)]
- **Cost** = Sum of all indicative cost values
- **Billings** = Sum of all indicative price values
- **Total cost** = Cost + Non-billable Expenses
- **Total Billings (Total Engagement Revenue or TER)** = Billings + Billable Expenses
- **Margin cost** = Total Billings – Total Cost
- **Margin percentage** = Margin $/Total Billings
- **EAF (Engagement Adjustment Factor)** = (Billings/NSR)-1
- **Blended rate** = Billings/Staffed Hours

6.6 Estimation computation explained

The estimate is computed on a workstream-by-workstream basis: each workstream effort is computed separately, with the total effort computed as the sum of the individual workstream estimate, plus the effort of the discover and project management estimates.

6.7 Linkage of conceptual computation to physical tables

One of the crucial features of the Estimator is the Object Inventory table in the Components worksheet, where users will see discernable impact of different object types, sizes and modes for different workstreams and their components. This table and its computations are based on information that is located in the Data_EE worksheet. This worksheet provides input on the workstreams, components and their specific object types. Once they are configured correctly, depending on size (small, medium or large) and mode (new or modified on), effort hours are estimated for certain phases such as define, design, build, test and deploy.

In order to accurately estimate the effort hours per workstream, stages or disciplines, each component, while they are in scope, in the Object Inventory table can be structured based on object type, mode and count. Users will be able to then reorganize their table by changing these drivers and observe different effort hours per workstreams. This helps users determine the more accurate estimate specific to their need and its impact across different tabs in the worksheet.

6.8 Explanation of basic mode

By default, when Estimator is in standard mode, all workstreams, stages and parameters will be in scope, which will eventually reflect on the overall financial estimate. However, when the current mode is changed to basic, project management workstream and discover stage along with the hypercare parameter switch to out of scope. Also, certain calculations and adjustments, such as complexity adjustment, geographic distribution adjustment, fixed effort calculation and contingency calculation, are disabled. For first-time users, the basic mode can help to eliminate the complexity driver and other drivers and help users understand the computation and calculations in a more simplistic manner.

6.9 Excel issues

One of the excel "glitches" that is noticeable in the workbook, especially in the Resources worksheet, is that in some cases, some of the column headers become invisible or just some part of the headers is visible. Users may scroll to the right or left of the page and notice the W1, Wn column headers not fully visible and that is due to how large the worksheet is.

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Document reference no. 2101-3682626
ED None

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